

PROJECT CONSTRAINTS											
WIND ZONE (NZS3604):	VERY HIGH										
DESIGN WIND SPEED:	50 m/s										
DESIGN WIND PRESSURE:	ULS = 1.50 kPa SLS = 1.05 kPa										
EARTHQUAKE ZONE (NZS3604):	ZONE 1										
SITE SUBGRADE:	STRUCTURE DESIGN IS IN ACCORDANCE WITH SOIL & ROCK CONSULTANTS INVESTIGATION REPORT, REFERENCE 240054, DATED 15 APRIL 2024. ALL EXCAVATIONS SHALL BE INSPECTED AND APPROVED BY A CHARTERED ENGINEER EXPERIENCED IN GEOTECHNICAL ENGINEERING AND FAMILIAR WITH ABOVE REPORT.										
MACROCLIMATE: (FOR STEEL DURABILITY AS PER AS/NZS 2312) (REFER DRAWING S0-1 FOR PROTECTIVE COATING NOTES, UNO)	C4										
STRUCTURAL STEEL QUALITY CONTROL (TO AS/NZS 5131)	ALL STRUCTURAL STEEL TO COMPLY WITH THE FOLLOWING CATEGORIES: <table><tr><th>CATEGORY TYPE</th><th>CATEGORY</th></tr><tr><td>IMPORTANCE LEVEL</td><td>IL2</td></tr><tr><td>CONSTRUCTION</td><td>CC2</td></tr><tr><td>SERVICE</td><td>SC1</td></tr><tr><td>FABRICATION</td><td>FC1</td></tr></table>	CATEGORY TYPE	CATEGORY	IMPORTANCE LEVEL	IL2	CONSTRUCTION	CC2	SERVICE	SC1	FABRICATION	FC1
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FABRICATION	FC1										

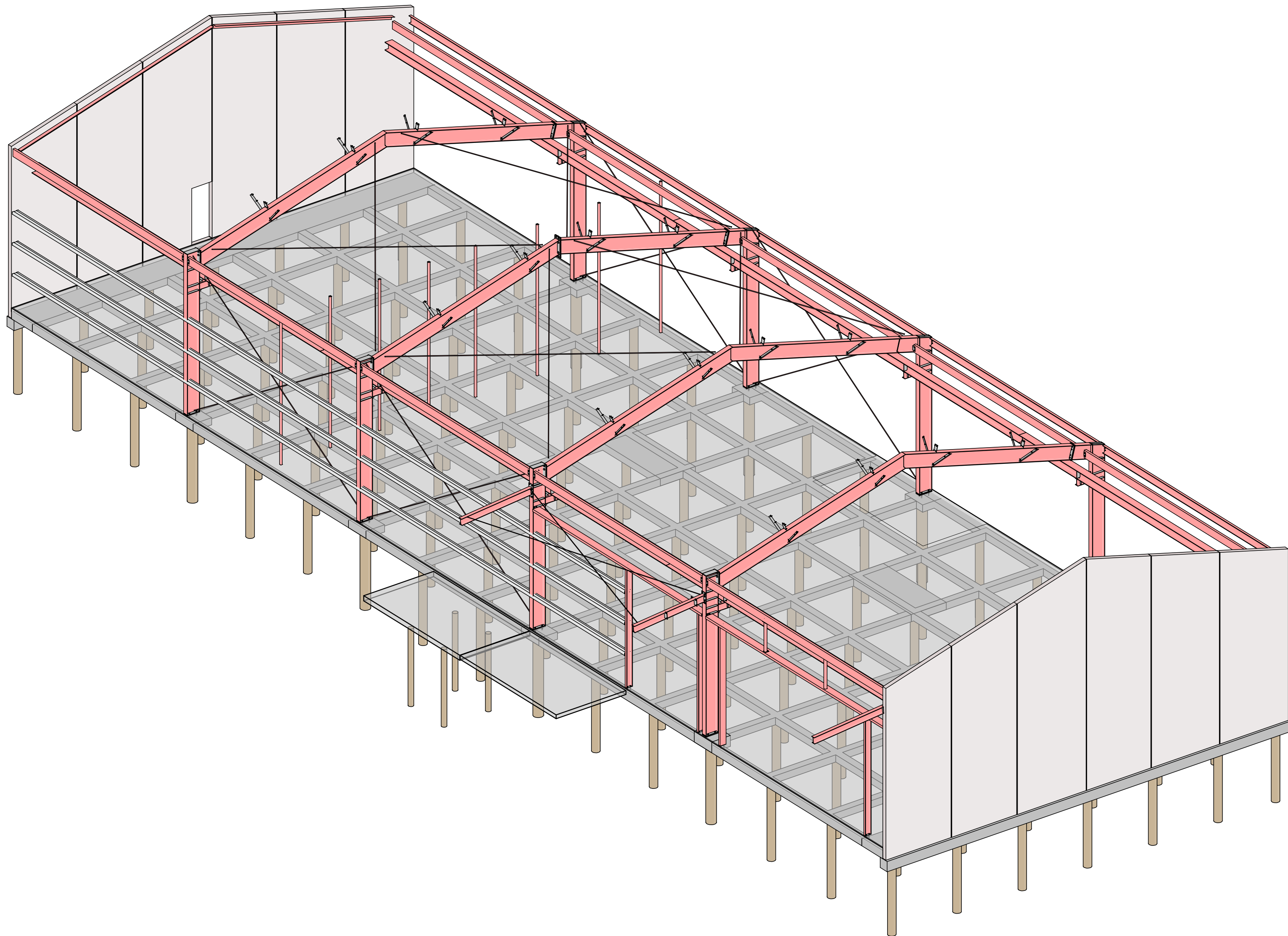
CONSTRUCTION OBSERVATION REQUIREMENTS		
COMPONENT	INSPECTION	REQUIRED
RECOMMENDED OVERALL LEVEL OF ENGINEERING SUPERVISION	WEEKLY INSPECTIONS AS A MINIMUM, OR AT KEY STAGES ONLY ON DISCUSSION WITH THE ENGINEER	CM
AUGERED RC PILES *	DEPTH, DIAMETER, SPACING, REINFORCING & COVER	N/A
CONCRETE ENCASED TIMBER POLES (INCLUDING RETAINING WALLS) *	DEPTH, DIAMETER, SPACING	N/A
DRIVEN STEEL PILES / TIMBER POLES	MEMBER SIZE, LENGTH, TARGET DEPTH, HAMMER, DROP, FINAL SET, DRIVING RECORD** (** TO BE SUPPLIED BY INSTALLER)	YES
RC FOOTINGS, PADS, PILE CAPS *	DEPTH, WIDTH, REINFORCING (INCLUDING STARTERS), COVER	YES
RCM FOUNDATION / RETAINING / SUPER-STRUCTURE WALLS	MORTAR WORKMANSHIP, WASH-OUTS, REINFORCING (INCLUDING STARTERS), COVER	YES
RIB RAFT GROUND FLOOR SLAB(S)	HARDFILL 1.0m BEYOND SLAB EDGE, REINFORCING, POD SET OUT, STARTERS, COVER, BOXING HEIGHT	YES
INSITU CONCRETE WALL(S)	THICKNESS, REINFORCING (INCLUDING STARTERS), COVER	N/A
SUSPENDED CONCRETE FLOOR(S)	REINFORCING (INCLUDING STARTERS), COVER	N/A
PRE-LINE STRUCTURAL STEEL	INSTALLATION, SIZE, CONNECTIONS, DURABILITY (IF EXTERNAL)	YES
PRE-LINE SPECIFICALLY DESIGN TIMBER STRUCTURE	INSTALLATION, SIZE, CONNECTIONS, DURABILITY (IF EXTERNAL)	YES

- NOTES:
- IN THE ABSENCE OF STRUCTURAL ENGINEERING INSPECTION REQUIREMENTS CONTAINED WITHIN THE BUILDING CONSENT APPROVAL THE ABOVE SHALL FORM THE MINIMUM LEVEL OF SUPERVISION REQUIRED.
 - OBSERVATIONS ARE TO BE UNDERTAKEN BY A SUITABLY QUALIFIED STRUCTURAL ENGINEER, FAMILIAR WITH THE DESIGN AND CONSENTED DOCUMENTATION.
 - A PRODUCER STATEMENT - CONSTRUCTION REVIEW (PS4) CERTIFICATE WILL ONLY BE ISSUED (BY THE ENGINEER ENGAGED) ON SATISFACTORY COMPLETION OF THE REQUIRED INSPECTIONS LISTED ABOVE.
 - 48 HOURS MINIMUM NOTICE IS REQUIRED FOR ANY INSPECTION.

* **LIMITATION:**
IN ALL INSTANCES THE STRUCTURAL ENGINEER IS NOT RESPONSIBLE FOR INSPECTION, TESTING OR THE LIKE OF SUBSOIL STRENGTH AND/OR CERTIFICATION OF HARDFILL COMPACTION. REFER ALL QUERIES OF THIS NATURE TO A CHARTERED GEOTECHNICAL ENGINEER.

WAREHOUSE EXTENSION

74 PATIKI RD, AVONDALE



DRAWING LIST		
SHEET	SHEET TITLE	REV
S0-0	COVER	A
S0-1	TYPICAL CONSTRUCTION NOTES	A
S0-2	TYPICAL CONSTRUCTION DETAILS	A
S0-3	ISOMETRIC VIEWS	A
S1-1	FOUNDATION & GROUND PLAN	A
S1-2	ROOF FRAMING PLAN	A
S2-1	SECTIONS & ELEVATIONS	A
S2-2	SECTIONS & ELEVATIONS	A
S2-3	SECTIONS & ELEVATIONS	A
S2-4	SECTIONS & ELEVATIONS	A
S3-1	FOUNDATION DETAILS	A
S3-2	FOUNDATION DETAILS	A
S3-3	PANEL DETAILS	A
S4-1	DETAILS	A
S4-2	DETAILS	A
S4-3	DETAILS	A
S4-4	DETAILS	A
TOTAL SHEETS: 17		

A - GENERAL

1. ALL STRUCTURAL DRAWINGS ARE TO BE READ IN CONJUNCTION WITH THE ARCHITECT'S PLANS, THE SPECIFICATION, AND ALL OTHER RELATED DOCUMENTS. THE NOTES PRESENTED WITHIN THESE SO SHEETS ARE GENERAL IN NATURE ONLY, AND A SITE SPECIFICATION SHALL TAKE PRECEDENCE OVER THESE NOTES.
2. REFER TO ARCHITECT'S DRAWINGS FOR SET-OUT AND LEVELS, UNO.
3. ANY DISCREPANCY SHALL BE REFERRED TO THE ENGINEER FOR COMMENT BEFORE PROCEEDING WITH THE WORK.
4. NO DIMENSIONS ARE TO BE OBTAINED FROM SCALING THE DRAWINGS.
5. UNLESS OTHERWISE NOTED, ALL LEVELS ARE IN METERS AND ALL DIMENSIONS ARE IN MILLIMETRES.
6. VERIFY ALL DIMENSIONS AND LEVELS PRIOR TO FABRICATION OR CONSTRUCTION OF ANY WORKS.

7. ALL WORKMANSHIP AND MATERIALS SHALL BE IN ACCORDANCE WITH THE REQUIREMENTS OF RELEVANT CURRENT NZS CODES AND RELATED BY-LAWS AND ORDINANCES OF LOCAL AND GOVERNMENT AUTHORITIES. (e.g. BUILDING CODE, BUILDING ACT).

DESIGN CODES.	
NZS 1170	GENERAL STRUCTURAL DESIGN AND LOADINGS FOR BUILDINGS
NZS AS 1720	TIMBER STRUCTURES
NZS 3101	THE DESIGN OF CONCRETE STRUCTURES
NZS 3404	THE DESIGN OF STEEL STRUCTURES
NZS 3604	LIGHT TIMBER FRAME BUILDINGS (NON-SPECIFIC DESIGN)
NZS 4229	NON-SPECIFIC DESIGN OF MASONRY BUILDINGS
NZS 4230	SPECIFIC DESIGN OF MASONRY STRUCTURES

MATERIALS AND WORKMANSHIP.	
NZS 3109	CONCRETE CONSTRUCTION
NZS 3114	FINISHES FOR CONCRETE
NZS 3121	WATER AND AGGREGATE FOR CONCRETE
NZS 3602	TIMBER AND WOOD PRODUCTS FOR USE IN BUILDING
NZS 3622	VERIFICATION OF TIMBER PROPERTIES
NZS 4210	MASONRY BUILDING
AS/NZS 1328	GLUED LAMINATED TIMBER
AS/NZS 1554	WELDING IN BUILDINGS
AS/NZS 2312	STRUCTURAL STEEL CORROSION PROTECTION
AS/NZS 4357	STRUCTURAL LAMINATED VENEER LUMBER
AS/NZS 4671	MILD STEEL DRAWN WIRE FOR CONCRETE REINFORCEMENT
AS/NZS 4671	STEEL FOR THE REINFORCEMENT OF CONCRETE
AS/NZS 5131	STRUCTURAL STEELWORK - FABRICATION AND ERECTION
SNZ TS 3404	DURABILITY REQUIREMENTS FOR STEEL STRUCTURES

8. THE CONTRACTOR SHALL BE RESPONSIBLE FOR MAINTAINING THE STRUCTURE AND NEIGHBOURING STRUCTURES IN A SAFE AND STABLE CONDITION DURING THE CONSTRUCTION PERIOD. NO PART SHALL BE OVER STRESSED.
9. CONTRACTOR IS RESPONSIBLE FOR ALL TEMPORARY WORKS NECESSARY TO COMPLETE THE CONSTRUCTION IN ACCORDANCE WITH THE DESIGN.
10. SERVICE PENETRATIONS MAY NOT HAVE BEEN LOCATED ON THE DRAWINGS. REFER TO ARCHITECT'S DRAWINGS (OR OTHERS) FOR FULL DETAILS. ANY PENETRATIONS NOT SHOWN ON STRUCTURAL DRAWINGS SHALL BE NOTIFIED TO ENGINEER BEFORE INSTALL.

11. WHERE PROPRIETARY PRODUCTS ARE SPECIFIED IN THE DOCUMENTS, THE CONTRACTOR MAY SUBMIT AN ALTERNATIVE PRODUCT TO THE ENGINEER FOR CONSIDERATION.

ABBREVIATIONS:

ALT	ALTERNATING	OD	OUTSIDE DIAMETER
CRS	CENTRES	PL	(STEEL) PLATE
COS	CONFIRM ON SITE	RCM	REINFORCED CONCRETE MASONRY
DIA	DIAMETER		
FL	FINISHED LEVEL	RL	REDUCED LEVEL
FGL	FINISHED GROUND LEVEL	SFL	STRUCTURAL FLOOR LEVEL
HDG	HOT DIP GALVANISED	SOP	SET OUT POINT
ID	INSIDE DIAMETER	SS	STAINLESS STEEL
MS	MILD STEEL	THK	THICK(NESS)
NTS	NOT TO SCALE	TYP	TYPICAL
O/A	OVERALL	TZS	THERMAL ZINC SPRAY
HORIZ	HORIZONTAL	UNO	UNLESS NOTED OTHERWISE
VERT	VERTICAL	¢	CENTERLINE

13. WHILE ALL REASONABLE CARE HAS BEEN TAKEN TO ADDRESS ALL STRUCTURAL ARRANGEMENTS, THE CONTRACTOR SHALL SEEK ADVICE FROM THE ENGINEER WHERE A DETAIL IS CONSIDERED MISSING OR INCORRECT.
14. FABRICATORS SHOP DRAWINGS WILL TYPICALLY REQUIRE A MINIMUM OF 5 WORKING DAYS TO REVIEW. THE CONTRACTOR SHALL ALLOW FOR THIS TIME IN THE WORKS PROGRAM.

B - STEELWORK

1. UNLESS STATED OTHERWISE THESE DRAWINGS SHOW ONLY THE DESIGN INTENT. SHOP DRAWINGS REMAIN THE RESPONSIBILITY OF THE CONTRACTOR.
2. ALL WORKMANSHIP AND MATERIALS TO BE IN ACCORDANCE WITH AS/NZS 5131:2016 AND RELATED DOCUMENTS.
3. ALL WELDS TO BE IN ACCORDANCE WITH AS/NZS 1554.1:2011 TO 1554.4:2014, AND RELATED STANDARDS.
4. APPROVED WASHERS, TAPERED WHERE NECESSARY, ARE TO BE USED UNDER ALL NUTS AND/OR TURNING BOLT HEADS.
5. UNLESS SHOWN OTHERWISE, ALL BASEPLATES SHALL BEAR DIRECTLY ON 20mm (±5mm) OF HIGH-STRENGTH LOW-SHRINK DRYPACK MORTAR (30MPa MINIMUM).
6. THE ENDS OF ALL HOLLOW SECTIONS SHALL BE CAPPED WITH A STEEL PLATE, THICKNESS TO MATCH SECTION WALL THICKNESS, UNO.
7. ALL CAPPED HOLLOW SECTIONS TO BE HOT DIP GALVANISED SHALL HAVE VENT HOLES (TO BE SHOWN ON THE SHOP DRAWINGS).
8. WELDED CONNECTIONS BETWEEN STRUCTURAL MEMBERS SHALL HAVE SP 6mm CONTINUOUS FILLET WELDS ALL SIDES, UNLESS NOTED OTHERWISE.
9. ALL COLD FORMED SECTIONS INCLUDING COLD ROLLED PURLINS TO CONFORM TO AS/NZS 4600:2018 AND SHALL BE ROLL-FORMED FROM ZINC COATED HIGH STRENGTH STEEL STRIP, ZINC-HI-TEN. MINIMUM YIELD STRESS 450 MPa, 300 g/m MINIMUM 2 COATING MASS, UNLESS NOTED OTHERWISE ON DRAWINGS.
10. PRIOR TO FABRICATION, THE STEEL FABRICATOR SHALL PROVIDE AN APPROPRIATE INSPECTION TEST PLAN (ITP) PER NZS 5131 TO THE ENGINEER. REFER SPECIFICATION FOR ANY SPECIAL REQUIREMENTS.
11. ALL CASES OF DAMAGE TO THE PROTECTIVE COATING OF STEELWORK SHALL BE BROUGHT TO THE ATTENTION OF THE ENGINEER AND REMEDIATED APPROPRIATELY.
12. FOR GALVANISED COATINGS, PATCH FIELD DAMAGE AND SITE WELDS WITH TWO COATS OF A TWO-PACK EPOXY POLYAMIDE ZINC-RICH PAINT CONTAINING A MINIMUM OF 92% ZINC DUST IN THE DRY FILM WHICH SHALL BE BUILT UP TO 75 MICROMETERS MINIMUM.

13.

STEEL ELEMENT GRADES UNO		
ELEMENT		GRADE
UB's, UC's, PFC's, TFC's, TFB's AND ANGLES		300
RHS, SHS, CHS		350
WB's, WC's		300
CONNECTION PLATES, STIFFENERS AND CLEATS		300
14. ALL STEEL CONNECTION PLATES, STIFFENERS AND CLEATS SHALL BE MINIMUM 8mm THICK, UNO.
15. PURLIN CLEATS ARE TO BE IN ACCORDANCE WITH THE MANUFACTURER'S STANDARD DETAILS, UNO.
16. CEILING SYSTEMS, DUCTWORK ETC. TO BE SUSPENDED FROM PURLINS SHOULD BE FIXED WITH HOOK BOLTS THROUGH PURLIN WEB. THE FLANGES OF THE PURLINS OR GIRTS SHALL NOT BE HOLED.
17. NO STEELWORK SHALL BE FABRICATED UNTIL ARCHITECT, ENGINEER, AND CONTRACTOR HAVE ACCEPTED SHOP DRAWINGS WITHOUT COMMENT.
18. BEAM CONNECTIONS ARE TO BE A MINIMUM OF A 10mm WEB PLATE WITH 2-M20 BOLTS (8.8/S) FOR SECTIONS 300mm DEEP AND LARGER, AND MIN 6mm WEB PLATE WITH 2-M16 BOLTS (8.8/S) FOR SECTIONS LESS THAN 300mm DEEP, UNO.

19. BOLT TYPES (AND DESIGNATIONS, WHERE USED) SHALL BE AS FOLLOWS:
 - 4.6/S
 - 8.8/S
 - 8.8/TB
 - 8.8/TF

20. M12 AND SMALLER BOLTS TO BE COMMERCIAL GRADE (4.6/S). M16 AND LARGER BOLTS TO BE HIGH STRENGTH SNUG TIGHTENED (8.8/S), UNO.
21. ALL CAST-IN HOLD-DOWN BOLTS SHALL BE GRADE 4.6/S, UNO. CONTRACTOR TO SET TO TEMPLATE AND CHECK LEVEL AND POSITION BEFORE CONCRETING.
22. STANDARD BOLT HOLES TO BE BOLT DIA +2mm FOR BOLTS SMALLER THAN OR EQUAL TO M24, AND +3mm FOR BOLTS LARGER THAN M24. HOLES IN BASE PLATES OVER CONCRETE MAY BE UP TO BOLT DIA +6mm WHERE 5mm PLATE WASHERS ARE USED.

STEEL PROTECTION:

- A. ALL STEEL PROTECTIVE COATINGS SHALL BE IN ACCORDANCE WITH AS/NZS 2312 AND SNZ TS 3404.
- B. EXCEPTIONS:
HOLLOW SECTIONS SHALL BE WELDED AIRTIGHT FOR TZS & IZS SYSTEMS. NO ONSITE DRILLING OF AIRTIGHT HOLLOW SECTIONS IS PERMITTED.
- C. ALL VISIBLE STEEL FINAL COATINGS SHALL BE IN ACCORDANCE WITH THE ARCHITECTURAL SPECIFICATIONS.
- D. RECOMMENDED PROTECTIVE COATINGS ARE AS FOLLOWS, UNO (REFER TO COVER SHEET FOR MACROCLIMATE):

C2 MACROCLIMATE		
LOCATION	SYSTEM DESIGNATION	AS/NZS 2312 YEAR TO FIRST MAJOR MAINTENANCE
INTERNAL - DRY (e.g. FLOOR)	IZS1	25+
INTERNAL - DAMP (e.g. EXTERNAL WALLS & ROOFS)	IZS4 HDG390 TZS100S	
EXTERNAL (e.g. SUBFLOOR, OPEN DECK)	IZS4 HDG390 TZS100S	
WET (e.g. CAST INTO GROUND / PILES)	HDG600-SD TZS150S	

C3 & C4 MACROCLIMATE		
LOCATION	SYSTEM DESIGNATION	AS/NZS 2312 YEAR TO FIRST MAJOR MAINTENANCE
INTERNAL - DRY (e.g. FLOOR)	IZS1	25+
INTERNAL - DAMP (e.g. EXTERNAL WALLS & ROOFS)	IZS4 HDG390 TZS100S	
EXTERNAL (e.g. SUBFLOOR, OPEN DECK)	HDG600-SD TZS200S	
WET (e.g. CAST INTO GROUND / PILES)	HDG600-SD TZS200S	

C5 MACROCLIMATE		
LOCATION	SYSTEM DESIGNATION	AS/NZS 2312 YEAR TO FIRST MAJOR MAINTENANCE
INTERNAL - DRY (e.g. FLOOR)	IZS1	25+
INTERNAL - DAMP (e.g. EXTERNAL WALLS & ROOFS)	IZS4 HDG600-4D TZS150S	15+
EXTERNAL (e.g. SUBFLOOR, OPEN DECK)	HDG600-SD TZS200S	25+
WET (e.g. CAST INTO GROUND / PILES)	HDG600-SD TZS200S	

- NOTES:
- a. THE TABLES ABOVE ARE GENERALLY FOR RESIDENTIAL CONSTRUCTION WHERE STRUCTURAL STEEL IS NOT ABLE TO BE EASILY ACCESSED FOR MAINTENANCE.
- b. WHERE STEEL IS EASILY ACCESSIBLE, OR REDUCED TIME TO FIRST MAINTENANCE IS ACCEPTABLE ON DISCUSSION WITH CLIENT, REFER ENGINEER AND FABRICATOR FOR RECOMMENDATIONS.
- c. ALTERNATIVE COATINGS ARE AVAILABLE - REFER TO ENGINEER FOR APPROVAL.
- d. COATINGS SELECTIONS SHALL BE MADE IN CONSULTATION WITH AN INTUMESCENT PAINT SUPPLIER WHERE REQUIRED
- e. IN ADDITION TO THE FINISHES SPECIFIED ABOVE, FOR INACCESSIBLE STEELWORK IN CONTACT WITH THE GROUND OR CAST INTO CONCRETE, WRAP IN GREASE IMPREGNATED TAPE OR EQUIVALENT 150mm ABOVE & BELOW THE GROUND/CONCRETE SURFACE, UNO.

24. AFTER TIGHTENING, EXPOSED FACES OF BOLTS, NUTS AND WASHERS SHALL BE PREPARED AND COATED AS SPECIFIED OR AS FOR ADJACENT STEELWORK.
25. DISSIMILAR METALS IN EXPOSED EXTERNAL STRUCTURAL STEEL,INCLUDING DIFFERENT GRADES OF STAINLESS STEEL (e.g. 304 AND 316) SHALL BE ISOLATED FROM ONE ANOTHER VIA AN APPROPRIATE BARRIER OR NON-CONDUCTIVE BUSH AND WASHER.

STEEL FIXINGS DURABILITY		
LOCATION		MATERIAL
DECKS		304SS OR 316SS
SHELTERED SUB-FLOOR AREAS	>600mm ABOVE FGL (INLAND)	HDG (BOLTS PASSING THROUGH POLES TO BE WELL GREASED)
	0-600mm ABOVE FGL OR COASTAL FRONTAGE	304SS OR 316SS
ROOF SPACES	ENCLOSED OR VENTILATED (INLAND)	HDG
	VENTILATED (COASTAL FRONTAGE)	304SS OR 316SS
CLOSED INTERNAL WALLS AND FLOORS		BLACK OR NOMINALLY GALVANISED
NOTE: STAINLESS STEEL FIXINGS THAT ARE EXPOSED FOR ARCHITECTURAL PURPOSES ARE RECOMMENDED TO BE GRADE 316.		

C - BLOCKWORK

1. ALL BLOCKWORK SHALL BE SOLID FILLED, UNO.
2. MAXIMUM COARSE AGGREGATE SIZE SHALL BE 12mm.
3. WALLS CAST USING HIGH LIFT METHOD REQUIRE CLEANOUT OPENINGS IN BOTTOM COURSE AT 0.4m TO 1.2m CENTRES MAX. DO NOT CLOSE UNTIL INSPECTED BY ENGINEER.
4. WHEN USING THE HIGH LIFT METHOD (LIFTS 1.2m TO MAX 2.4m TALL), SIKA CAVEY ADMIXTURE OR EQUIVALENT SHALL BE APPLIED TO BLOCK GROUT AT A RATE OF 1kg / m³, AND AVOID VIBRATING OR ROD COMPACTING THE GROUT.
5. ALL MASONRY UNITS SHALL BE MANUFACTURED IN ACCORDANCE WITH AS/NZS 4455.1:2008. WALLS SHALL BE CONSTRUCTED TO THE REQUIREMENT OF NZS 4210:2001 FOR GRADE B MASONRY, UNO.
6. ALL WALLS TO BE REINFORCED WITH D12 BARS AT 600crs EACH WAY, UNO.
7. VERTICAL CONTROL JOINTS SHALL BE CONSTRUCTED AT 4800mm MAX CENTRES. PROVIDE VERTICAL REINFORCEMENT IN CELLS ADJACENT TO JOINT (REFER DETAIL ON SHEET S0-2).
8. CONCRETE MASONRY BLOCKS SHALL HAVE A COMPRESSIVE STRENGTH OF 12MPa (GRADE B) AND CONFORM TO NZS 4210, UNO. DENSITY SHALL BE > 1750 kg/m.
9. MORTAR: MORTAR SHALL COMPLY WITH THE REQUIREMENTS OF NZS 4210 WITH NOMINAL PROPORTIONS 1 PART CEMENT, 3 PARTS SAND AND 1/4 PART LIME UNLESS NOTED OTHERWISE. COMpressive STRENGTH OF MORTAR SHALL BE NOT LESS THAN 20 MPa.
10. GROUT: GROUT SHALL HAVE A SLUMP OF 230mm AND COMPRESSIVE STRENGTH f_m=20 MPa (25 MPa IN SEA SPRAY ZONE).
11. MORTAR JOINTS TO BE 10mm THICK WITH BLOCKS FULLY BEDDED AND PERPENDICULAR ENDS FILLED. JOINTS TO BE TOOLED AT EXPOSED OR RENDERED SURFACES. BEFORE PLACING VERTICAL REINFORCEMENT, CORES AND BARS TO BE CLEANED OF ALL MORTAR FINES AND DROPPINGS.
12. CHASING REQUIRED IN LOAD BEARING MASONRY UNITS TO BE APPROVED BY DESIGN ENGINEER.
13. ALL GROUTING AND PLACEMENT OF MASONRY UNITS SHALL BE BY A REGISTERED MASON OR A COMPETENT PERSON SUPERVISED DIRECTLY BY A REGISTERED MASON.
14. COVER AND REINFORCING ALIGNMENTS SHALL BE AS PER THE DRAWN DETAILS. MISALIGNED REINFORCING SHALL BE REPLACED TO THE ENGINEER'S SATISFACTION.
15. THE CONTRACTOR SHALL ADVISE THE ENGINEER AT LEAST 48 HOURS PRIOR TO GROUTING BLOCKWORK TO ENABLE THEM TO INSPECT THE WORK.
16. UNLESS NOTED OTHERWISE ON THE DRAWINGS, THE FOLLOWING DEFORMED BAR SPlice LENGTHS SHALL BE USED:

SPlicing OF MASONRY REINFORCEMENT				
BAR DIAMETER	VERT.	HORIZ.	VERT.	HORIZ.
10	400	620	700	910
12	480	820	840	1100
16	640	830	1120	1460
20	800	1040	1400	1820
NOTE: HORIZONTAL LAP LENGTHS HAVE BEEN INCREASED PER NZS 4230 C6.3.7.5(a)				

D - CONCRETE

1. ALL CONCRETE CONSTRUCTION TO BE IN ACCORDANCE WITH NZS 3109:1997 AND RELATED DOCUMENTS.

CONCRETE PROPERTIES, UNO			
LOCATION	f _c AT 28 DAYS	SLUMP	AGGREGATE SIZE
FOUNDATIONS	30 MPa	100mm TYP	19mm, UNLESS AGREED OTHERWISE WITH ENGINEER
SLABS	30 MPa		
COLUMNS/BEAMS	30 MPa		
PRECAST ELEMENTS	40 MPa		

3. UNLESS NOTED OTHERWISE ON THE DRAWINGS, THE FOLLOWING DEFORMED BAR SPlice LENGTHS SHALL BE USED (30MPa CONCRETE):

SPlicing OF CONCRETE REINFORCEMENT		
BAR DIAMETER	GRADE 300 (D)	GRADE 500 (H)
10	400	600
12	450	750
16	600	950
20	750	1250
25	1000	1500
32	1150	1900

- NOTE:
- A. PLAIN ROUND BARS SHALL NOT BE USED OTHER THAN FOR STIRRUPS, TIES, SPIRAL AND LINKS.
- B. SPlice LENGTHS ABOVE ASSUME BARS IN FULL CONTACT AND NO MORE THAN 3 BARS IN A GROUP.
- C. IF MORE THAN 50% OF BEAM BARS ARE LAPPED AT ONE LOCATION, SPlice LENGTHS SHOWN ABOVE SHALL BE INCREASED BY 30%.
- D. NO REINFORCEMENT SPlices SHALL BE MADE, OTHER THAN THOSE SHOWN ON THE STRUCTURAL DRAWINGS, WITHOUT THE PRIOR APPROVAL OF THE ENGINEER. MINIMUM LAP FOR FABRIC SHALL BE ONE MESH GRID PLUS 25mm.

4. PREFABRICATED WIRE MESH REINFORCING SHEETS (e.g. SE62) SHALL BE LAID WITH A MINIMUM OF ONE OVERLAPPING SQUARE BETWEEN SHEETS OR AS PER SUPPLIER'S LITERATURE.
5. THE FOLLOWING REINFORCEMENT COVER SHALL BE USED, UNO:

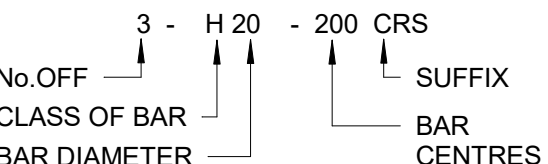
REINFORCEMENT COVER (30MPa CONCRETE)			
LOCATION	EXAMPLES	EXPOSURE OR CLIMATE	MINIMUM COVER
INTERIOR	TOP SURFACE OF INTERNAL SLAB, COLUMNS, BEAMS	DRY	25
EXTERIOR	EXTERIOR SLAB, PRECAST OR IN-SITU WALL EXTERIOR FACE	INLAND	30
		COASTAL PERIMETER	35
IN CONTACT WITH GROUND	FOOTINGS WITH DPM UNDER	CAST AGAINST DPM, FORM, OR SITE CONCRETE	50
	PILES	NOT CAST AGAINST DPM	75
NOTE: A. MINIMUM COVER IS TO OUTERMOST REINFORCEMENT. B. COVER SHALL BE MAINTAINED BY THE USE OF APPROVED BAR CHAIRS AT 1200crs FOR BARS UP TO 16 DIA AND 2000crs FOR BARS 20 DIA AND LARGER.			

6. CEMENT USED IN THE CONCRETE SHALL CONFORM TO THE FOLLOWING:
 - ORDINARY PORTLAND CEMENT TO NZS 3122:2009.
 - HIGH GRADE CEMENT TO NZS 3104:2021.
7. DIMENSIONS OF CONCRETE ELEMENTS DO NOT INCLUDE THICKNESS OF APPLIED FINISHES (e.g. SCREED), UNO.
8. BEAM DEPTHS ARE NOTED FIRST (I.e. D x W) AND GENERALLY INCLUDE THE THICKNESS OF SLAB, IF ANY.
9. CONCRETE WALLS AND ON-GRADE FLOORS REQUIRE CONSTRUCTION JOINTS. CONSTRUCTION JOINTS WHERE NOT SHOWN ON DRAWINGS SHALL BE 'TYPE B', AND LOCATED TO THE APPROVAL OF THE ENGINEER AND ARCHITECT.
10. ALL FORMWORK AND PROPPING WORKS SHALL BE THE CONTRACTOR'S RESPONSIBILITY.
11. NO PENETRATIONS, CHASES OR EMBEDMENTS OF PIPES OTHER THAN THOSE SHOWN ON THE STRUCTURAL DRAWINGS SHALL BE MADE IN CONCRETE MEMBERS WITHOUT THE PRIOR APPROVAL OF THE ENGINEER.
12. REINFORCEMENT IS REPRESENTED DIAGRAMMATICALLY AND NOT NECESSARILY IN TRUE PROJECTION.
13. ALL CHASES, CONDUITS, PIPES, AND REBATES NOT SHOWN WITHIN THESE DRAWINGS REQUIRE ENGINEERS APPROVAL BEFORE CONSTRUCTION.
14. CAMBERS UNLESS NOTED OTHERWISE ON THE DRAWINGS AND EXCEPT FOR PRESTRESSED WORK SHALL BE PROVIDED IN BEAMS AND SLABS AS FOLLOWS:
 - SPANS GENERALLY = 0.002 x SPAN
 - CANTILEVERS = 0.004 x CANTILEVER LENGTH
15. FINISHES TO CONCRETE TO BE AS FOLLOWS (ALL IN ACCORDANCE WITH NZS 3114) UNO ON DRAWINGS OR SPECIFIED BY ARCHITECT:
 - TOPS OF SLABS, RAMPS-INTERNAL. = U3 POWER FLOAT.
 - TOPS OF SLABS, RAMPS-EXTERNAL. = U3 LIGHT BROOM (NYLON)
16. NO PRECAST SHALL BE FORMED UNTIL ARCHITECT, ENGINEER, AND CONTRACTOR HAVE ACCEPTED SHOP DRAWINGS, INCLUDING REINFORCEMENT, WITHOUT COMMENT.
17. ANY PROPRIETARY COMPONENTS SHALL BE INSTALLED AS PER THE SUPPLIER'S INSTRUCTIONS AND A COPY OF THE DOCUMENTATION SUBMITTED TO THE ENGINEER, ARCHITECT AND MAIN CONTRACTOR FOR APPROVAL.
18. PRECAST CONCRETE TOLERANCES PER NZS 3109 TABLE 5.1:
 - DIMENSIONAL = +5mm.
 - STRAIGHTNESS = 1 IN 1000
19. LIGHTWEIGHT CONCRETE SCREED DENSITY SHALL BE < 1250 kg/m³.
20. UNDER-SLAB POLYSTYRENE TO BE 20 kg/m³, UNO, AND ISOLATED FROM GROUND WITH A DPM (REFER ARCHITECT FOR WATER PROOFING).

E - CONCRETE AND RCM REINFORCING

1. MAX 100mm THICK SLAB-ON-GRADE OR TOPPING SHALL BE REINFORCED WITH SE62 DUCTILE MESH, UNO.
2. NO WELDING OF BARS PERMITTED, UNO.
3. RE-BENDING OF SEISMIC GRADE 500 MA BARS PERMITTED UNDER ENGINEER'S SUPERVISION ONLY.
4. CONDUITS IN SLABS ARE TO BE PLACED ABOVE BOTTOM REINFORCEMENT AND BELOW TOP REINFORCEMENT AND SPACED OFF REINFORCING BY MINIMUM COVER REQUIREMENTS SHOWN IN TABLE D5.
5. HOOK LAPS AND BENDS TO BE IN ACCORDANCE WITH NZS 3109, UNO. REFER TO S0-2 FOR EXAMPLES.
6. CAST-IN BARS SHALL NOT BE BENT TO CORRECT A MISALIGNMENT. REFER ENGINEER FOR REMEDIAL DETAIL.
7. BARS SHALL BE CLEANED OF LOOSE RUST SCALE, GROUT, AND DIRT BEFORE POUR.

REINFORCEMENT NOTATION:



BAR CLASSES:			
R	PLAIN ROUND	F _y =	300 MPa
HR	PLAIN ROUND	F _y =	500 MPa
D	DEFORMED	F _y =	300 MPa
H	DEFORMED	F _y =	500 MPa
SUFFIX:			
T	TOP	B	BOTTOM
EF	EACH FACE	FF	FAR FACE
EW	EACH WAY	C	CENTRAL
NF	NEAR FACE	CRS	CENTRES

F - TIMBER

1. ALL SAWN TIMBER IDENTIFIED ON THE STRUCTURAL DRAWINGS IS TO BE SEASONED SG 8 (UNO) AND COMPLY WITH THE REQUIREMENTS OF NZS 3622 AND NZS AS 1720.
2. THE MINIMUM LEVEL OF TIMBER TREATMENT AND PROTECTION IS TO BE IN ACCORDANCE WITH NZS 3602 AND AS OUTLINED IN NZS3604 - SECTION 4 DURABILITY, UNO.
3. FRAMING AND STRUCTURAL TIMBER SHALL BE SEPARATED FROM CONCRETE OR CONCRETE MASONRY IN ACCORDANCE WITH 2.3.3 OF NZS 3604.
4. UNO, USE GRADE H5 WHERE TIMBER IS IN DIRECT CONTACT WITH GROUND, GRADE H4 WHERE EXPOSED TO WEATHER, AND H1 TREATED WHERE FULLY ENCLOSED.

5. UNLESS NOTED OTHERWISE, A MINIMUM LEVEL OF CONNECTIVITY BETWEEN TIMBER MEMBERS IS EXPECTED IN ACCORDANCE WITH NZS 3604.
6. THE MAXIMUM MOISTURE CONTENT PERMITTED IS 16%, UNLESS THE TIMBER HAS SPECIFICALLY BEEN DESIGNED FOR WET EXTERNAL USE. TIMBER THAT HAS A HIGHER MOISTURE CONTENT THAN THIS LIMIT MUST BE PROPPED AND KEPT DRY UNTIL THE MOISTURE CONTENT HAS RETURNED TO LESS THAN 25%. THE CONTRACTOR IS RESPONSIBLE FOR TIMBER MOISTURE MANAGEMENT DURING CONSTRUCTION.
7. STEEL FASTENERS DETAILED ARE GENERALLY FROM MITEK. ALTERNATIVE PRODUCTS MAY BE USED WITH THE APPROVAL OF THE ENGINEER.
8. NO DEFECTS TO OCCUR WITHIN 150mm OF ANY CONNECTION.
9. WHERE TIMBER TRUSSES HAVE BEEN SHOWN ON THE DRAWINGS, THE CONTRACTOR SHALL ENGAGE A TRUSS MANUFACTURER TO DESIGN THE TRUSSES, UNLESS ALREADY PROVIDED. SHOP DRAWINGS SHALL BE SUBMITTED TO THE MAIN CONTRACTOR, ARCHITECT, AND ENGINEER FOR COMMENT PRIOR TO FABRICATION.
10. ALL BOLTS AND NUTS TO HAVE WASHERS UNDER HEAD AND NUT (OTHER THAN A COACH BOLT HEAD).
11. ALL BOLTS TO BE RETIGHTENED PRIOR TO COVERING OR LEAVING JOB.
12.

MINIMUM WASHER DIMENSIONS, UNO.		
BOLT DIAMETER	SQUARE WASHER (mm)	ROUND WASHER (mm)
M6	25 x 25 x 1.6	30 x 1.6
M8	32 x 32 x 2.0	36 x 2.0
M10	40 x 40 x 2.5	45 x 2.5
M12	50 x 50 x 3.0	55 x 3.0
M16	60 x 60 x 4.0	65 x 4.0
M20	65 x 65 x 5.0	75 x 5.0
>M20	75 x 75 x 6.0	85 x 6.0

G - DESIGN LOADS

1. LOADS TO SUSPENDED CONCRETE FLOORS ARE AS SHOWN ON PLAN. IF NONE PRESENT, REQUEST FROM ENGINEER.
2. ULTIMATE AND SERVICEABILITY SITE WIND PRESSURES ARE AS SHOWN ON COVER SHEET.

H - FOUNDATIONS

1. 40mm OF SITE CONCRETE (10 MPa) IS RECOMMENDED UNDER ALL FOUNDATIONS IF ADVERSE WEATHER FORECAST.
2. CONTRACTOR TO ALLOW TO PROP EXCAVATIONS IN NON-COHESIVE SUBGRADE (e.g. SAND OR GLACIAL TILL) TO PREVENT COLLAPSES DURING WORKS.
3. FOOTING LEVELS, WHERE SHOWN, ARE ESTIMATES ONLY, AND SHALL BE VERIFIED BY THE CONTRACTOR ON SITE.
4. SUBGRADE:
REFER TO COVER SHEET FOR DESIGN ASSUMPTIONS.
5. BASECOURSE:
BASECOURSE SHALL BE GAP40, COMPACTED IN 150mm LAYERS WITH VIBRATING PLATE COMPACTOR. FILL DEPTHS MORE THAN 600mm TO BE TESTED BY A GEOTECHNICAL ENGINEER.

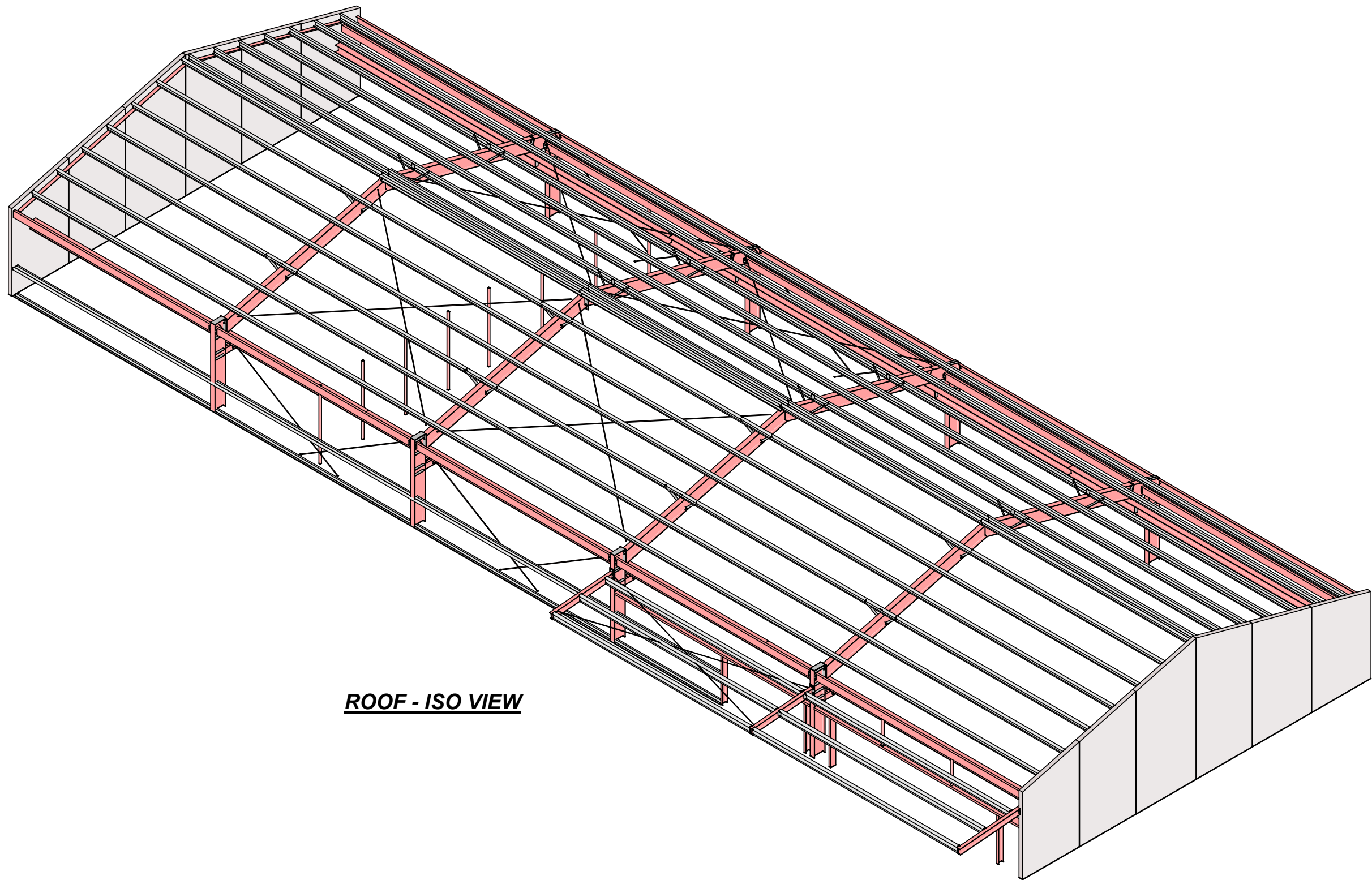
J - CHEMICAL ANCHORING

"INJECTION ANCHORS" ARE TO BE ONE OF THE FOLLOWING ACCEPTED EPOXY SYSTEMS:

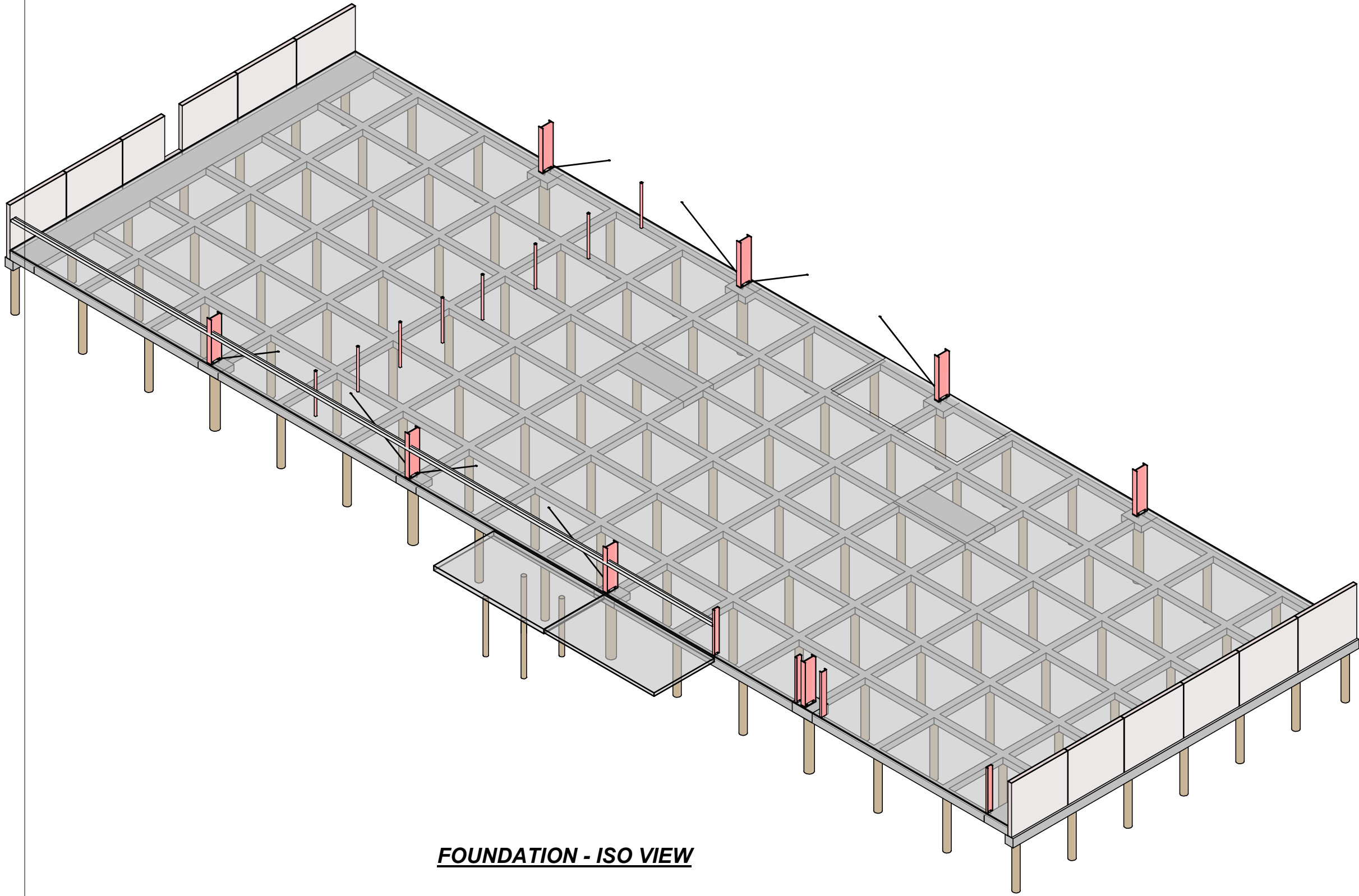
- HLTI HIT-HY 200;
- TECFI SWE01;
- RAMSET EPOC8 C8 XTREM.

INSTALLATION DEPTH & TIGHTENING TORQUE TO MANUFACTURER'S RECOMMENDATIONS, UNO.

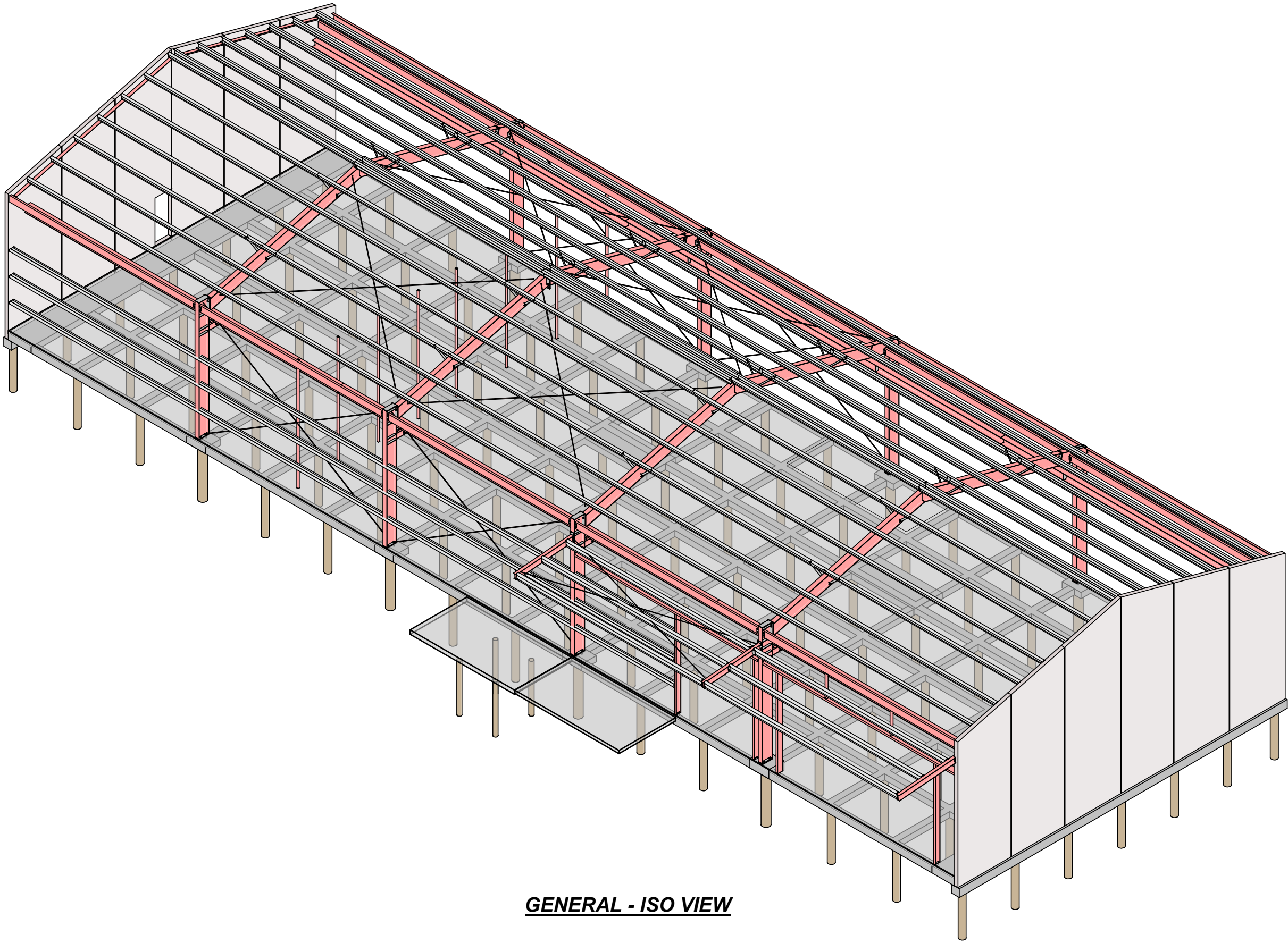
A	2025.08.11	BUILDING CONSENT
Rev	Date	Description
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Project		
WAREHOUSE EXTENSION		
74 PATIKI RD, AVONDALE		
Drawing Title		
TYPICAL CONSTRUCTION NOTES		
Engineer: AF	Drawn: DT	Date: MAY'25
Checked: AF	Drg Size: A1	Scales: NTS
Job No: SHL6538	Drg. No: S0-1	Rev: A



ROOF - ISO VIEW



FOUNDATION - ISO VIEW



GENERAL - ISO VIEW

STRUCTURAL LEGEND	
STRUCTURAL FRAMING	
GB1	300W x 420D 30MPa CONCRETE GROUND BEAM. REINFORCED WITH 3-H16 TOP & BOT PLUS ADDITIONAL 2-H16 x 2.0m LONG OVER ALL PILE AND RH10 STIRRUPS AT 150 CRS, 50mm TYP COVER
GB2	400W x 400D 30MPa CONCRETE GROUND BEAM. REINFORCED WITH 4-H16 TOP & BOT AND RH10 STIRRUPS AT 150 CRS, 75mm TYP COVER
R1	DHS250/15 WITH 2 SET OF BRACE AT 1200 CRS
R2	DHS300/18 AT 1200 CRS
RB1	530UB93
RB2	300PFC
RB3	250UB37
RB4	150x10 EA
RB16 ROD	16 DIA REIDBRACE ROD
RB25 ROD	25DIA REIDBRACE ROD
SB1	310UC118
SB2	250UB37
SB3	300 PFC ON FLAT

NOTES	
1.	REFER TO COVER PAGE FOR MANDATORY ENGINEER INSPECTIONS.
2.	REFER TO COVER PAGE & SHEET S0-1 FOR STEEL PROTECTION COATING REQUIREMENTS.
3.	REFER TO SHEETS S0-1 & S0-2 FOR TYPICAL CONSTRUCTION NOTES & DETAILS.

STRUCTURAL LEGEND	
STRUCTURAL FOUNDATION	
P1	300 SED H5 'UGLY' TANAPOLE DRIVEN TO REFUSAL, 8.0m MINIMUM BELOW GROUND BEAM INVERT. TEST PILES ARE REQUIRED TO VERIFY POLE LENGTH. 10mm SET PER BLOW FOR 5 CONSECUTIVE BLOWS (HAMMER 500 kG, DROP 1.0m)
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P3	200 SED H5 'UGLY' TANAPOLE DRIVEN TO FINAL SET 15mm, 6.0m MINIMUM BELOW GROUND BEAM INVERT. TEST PILES ARE REQUIRED TO VERIFY POLE LENGTH (HAMMER 500 kG, DROP 1.0m)
PAD 1	1000x800x420D 30MPa CONCRETE PAD. REINFORCED WITH H12-150 CRS EW TOP & BOT
PAD 2	3600x1500x420D 30MPa CONCRETE PAD. REINFORCED WITH H16-150 CRS TOP & BOT AND 6 LEGS R10-150 CRS, 75mm TYP COVER
STRUCTURAL WALL	
PW1	175 THK 40MPa CONCRETE PRECAST WALL. REINFORCED WITH D12-100 CRS VERTICAL AND D12-250 CRS HORIZONTAL
STRUCTURAL SLAB	
S1	140mm THK 30MPa CONCRETE RAFT SLAB REINFORCED WITH H12-150 CRS EW TOP AND H12-150 CRS BOT ACROSS FLOOR (SHORT SPAN) AND H12-300 CRS BOT ALONG SLAB (LONG SPAN)
S2	200mm THK 30MPa CONCRETE SLAB ON GRADE. REINFORCED WITH H12-200 CRS EW TOP AND H12-200 CRS EW BOTTOM, ON SAND BLINDING ON 100 MIN COMPACTED HARDFILL
S3	150mm THK 30MPa CONCRETE SLAB ON GRADE. REINFORCED WITH SE92 MESH, ON SAND BLINDING ON 100 MIN COMPACTED HARDFILL
STRUCTURAL COLUMN	
SP1	530UB93
SP2	250 PFC
SP3	89x6 SHS POST
SP4	75x6 SHS

A	2025.08.11	BUILDING CONSENT
Rev	Date	Description
REVISIONS		

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Sullivan Hall

CHARTERED ENGINEERS

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Project

WAREHOUSE EXTENSION

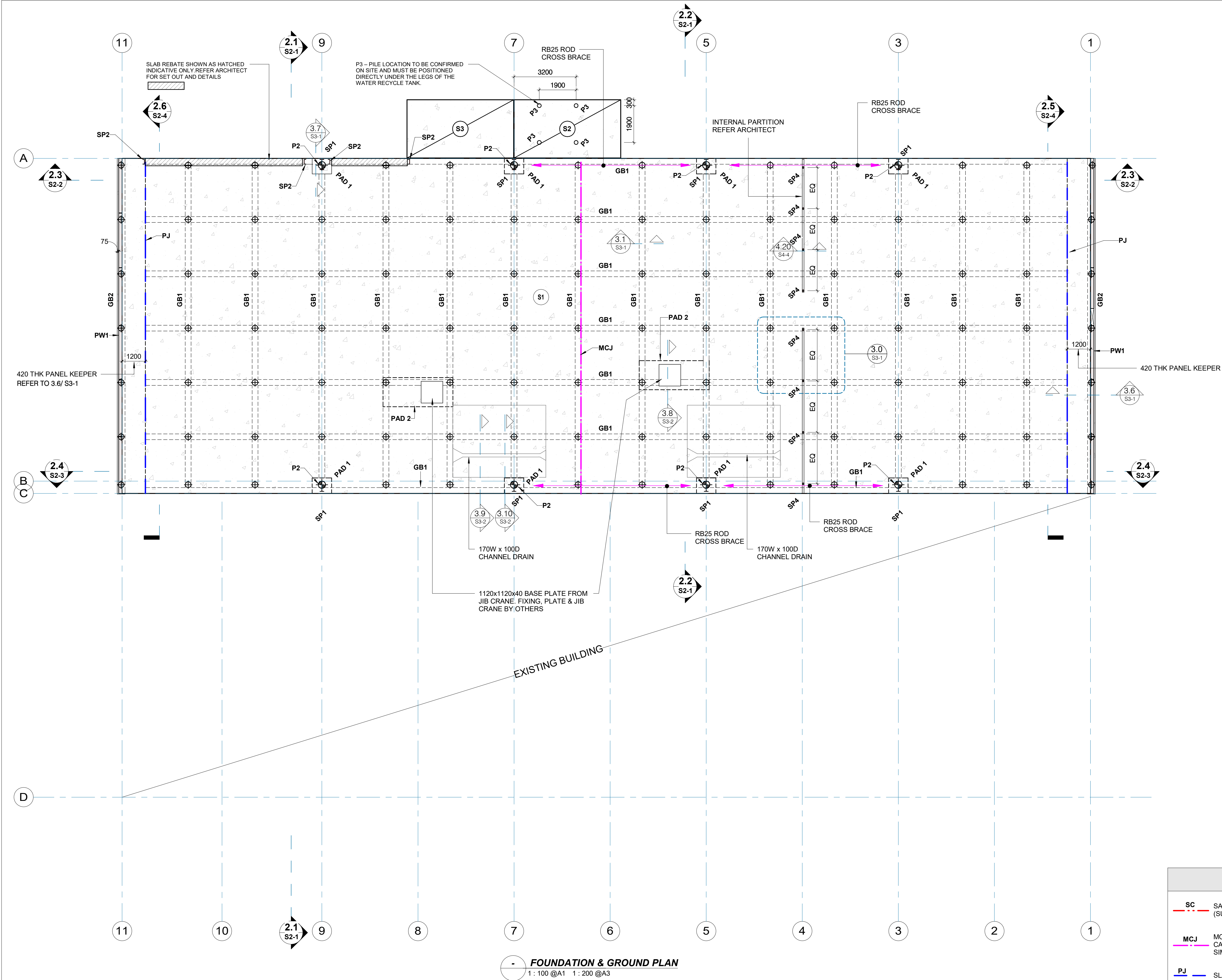
74 PATIKI RD, AVONDALE

Drawing Title

ISOMETRIC VIEWS

Engineer: AF	Drawn: DT	Date: MAY'25
Checked: AF	Drg Size: A1	Scales: NTS
Job No.: SHL6538	Drg. No.: S0-3	Rev: A

11/08/2025 1:14:28 PM



FOUNDATION & GROUND PLAN
1 : 100 @A1 1 : 200 @A3

LEGEND		
SC	SAW CUT (SUSPENDED SLAB NOT REQUIRED)	
MCJ	MOVEMENT CONSTRUCTION JOINT. CANZAC SPEED DOWEL SYSTEM OR SIMILAR.REFER SHEET S0-2	
PJ	SLAB POUR JOINT.	
	P1 PILES. REFER SCHEDULE	
	P2 PILES. REFER SCHEDULE	
	P3 PILES. REFER SCHEDULE	

WAREHOUSE CONCRETE SLAB LOADING:
Q = 24KPa.

- NOTES
1.

REFER TO COVER PAGE FOR MANDATORY ENGINEER INSPECTIONS.
2.

REFER TO COVER PAGE & SHEET S0-1 FOR STEEL PROTECTION COATING REQUIREMENTS.
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STRUCTURAL LEGEND	
STRUCTURAL FOUNDATION	
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PAD 2	3600x1500x420D 30MPa CONCRETE PAD. REINFORCED WITH H16-150 CRS TOP & BOT AND 6 LEGS R10-150 CRS, 75mm TYP COVER
STRUCTURAL WALL	
PW1	175 THK 40MPa CONCRETE PRECAST WALL. REINFORCED WITH D12-100 CRS VERTICAL AND D12-250 CRS HORIZONTAL
STRUCTURAL SLAB	
S1	140mm THK 30MPa CONCRETE RAFT SLAB REINFORCED WITH H12-150 CRS EW TOP AND H12-150 CRS BOT ACROSS FLOOR (SHORT SPAN) AND H12-300 CRS BOT ALONG SLAB (LONG SPAN)
S2	200mm THK 30MPa CONCRETE SLAB ON GRADE. REINFORCED WITH H12-200 CRS EW TOP AND H12-200 CRS EW BOTTOM, ON SAND BLINDING ON 100 MIN COMPACTED HARDFILL
S3	150mm THK 30MPa CONCRETE SLAB ON GRADE. REINFORCED WITH SE92 MESH, ON SAND BLINDING ON 100 MIN COMPACTED HARDFILL
STRUCTURAL COLUMN	
SP1	530UB93
SP2	250 PFC
SP4	75x6 SHS
STRUCTURAL FRAMING	
GB1	300W x 420D 30MPa CONCRETE GROUND BEAM. REINFORCED WITH 3-H16 TOP & BOT PLUS ADDITIONAL 2-H16 x 2.0m LONG OVER ALL PILE AND RH10 STIRRUPS AT 150 CRS, 50mm TYP COVER
GB2	400W x 400D 30MPa CONCRETE GROUND BEAM. REINFORCED WITH 4-H16 TOP & BOT AND RH10 STIRRUPS AT 150 CRS, 75mm TYP COVER

Rev	Date	Description
A	2025.08.11	BUILDING CONSENT
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REVISIONS		

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Project

WAREHOUSE EXTENSION

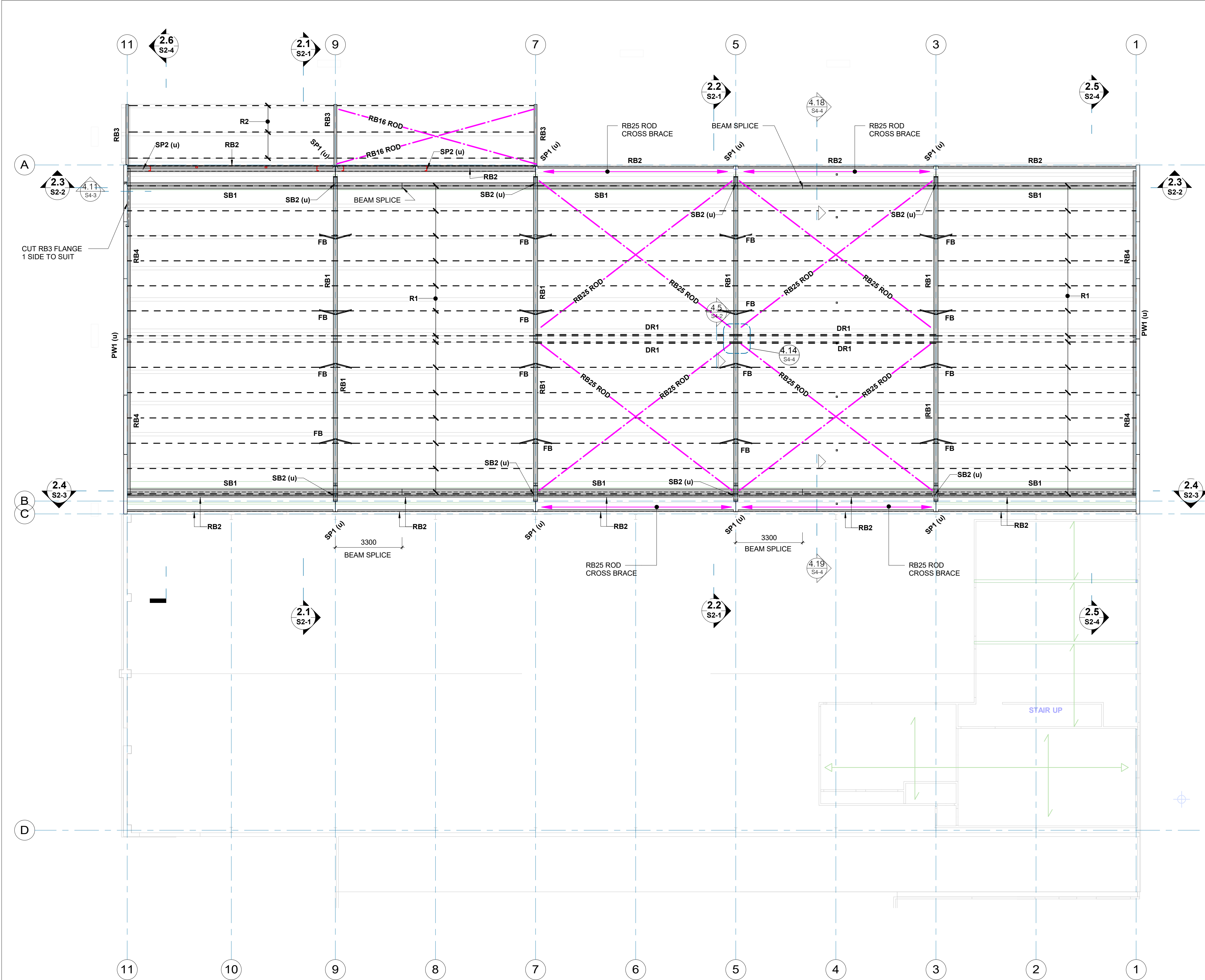
74 PATIKI RD, AVONDALE

Drawing Title
FOUNDATION & GROUND PLAN

Engineer: **AF** Drawn: **DT** Date: **MAY'25**

Checked: **AF** Drg Size: **A1** Scales: **As indicated**

Job No.: **SHL6538** Drg. No.: **S1-1** Rev: **A**



ROOF FRAMING PLAN
1: 100 @A1 1: 200 @A3

NOTES

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- REFER TO COVER PAGE & SHEET S0-1 FOR STEEL PROTECTION COATING REQUIREMENTS.
- REFER TO SHEETS S0-1 & S0-2 FOR TYPICAL CONSTRUCTION NOTES & DETAILS.

STRUCTURAL LEGEND

STRUCTURAL WALL	
PW1	175 THK 40MPa CONCRETE PRECAST WALL. REINFORCED WITH D12-100 CRS VERTICAL AND D12-250 CRS HORIZONTAL

STRUCTURAL COLUMN	
SP1	530UB93
SP2	250 PFC
SP3	89x6 SHS POST
SP4	75x6 SHS

STRUCTURAL FRAMING	
R1	DHS250/15 WITH 2 SET OF BRACE AT 1200 CRS
R2	DHS300/18 AT 1200 CRS
RB1	530UB93
RB2	300PFC
RB3	250UB37
RB4	150x10 EA
RB16 ROD	16 DIA REIDBRACE ROD
RB25 ROD	25DIA REIDBRACE ROD
SB1	310UC118
SB2	250UB37
SB3	300 PFC ON FLAT

LEGEND

- (u) DENOTES UNDER
- FB FLY BRACE

Rev	Date	Description
A	2025.08.11	BUILDING CONSENT

REVISIONS

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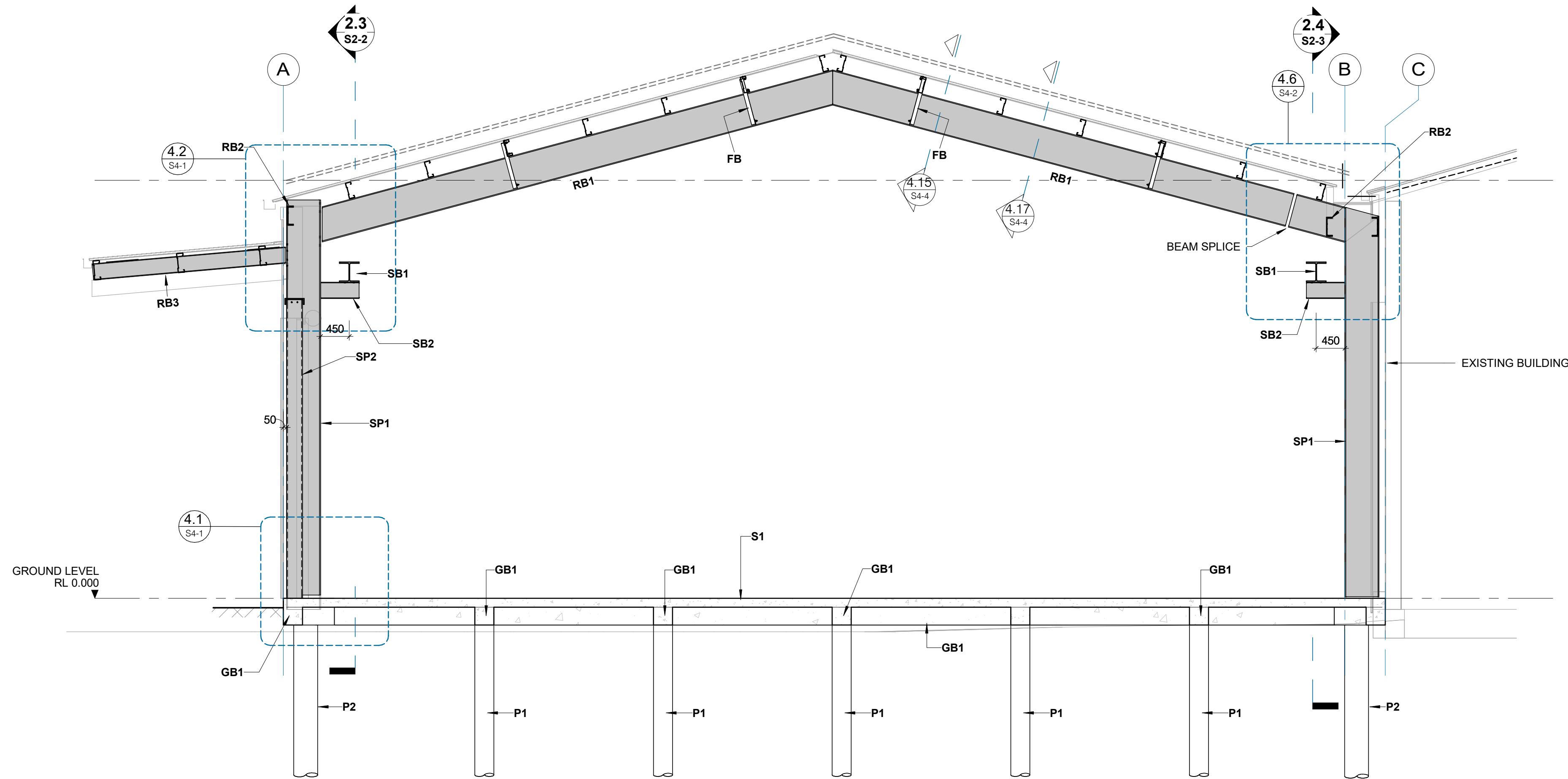
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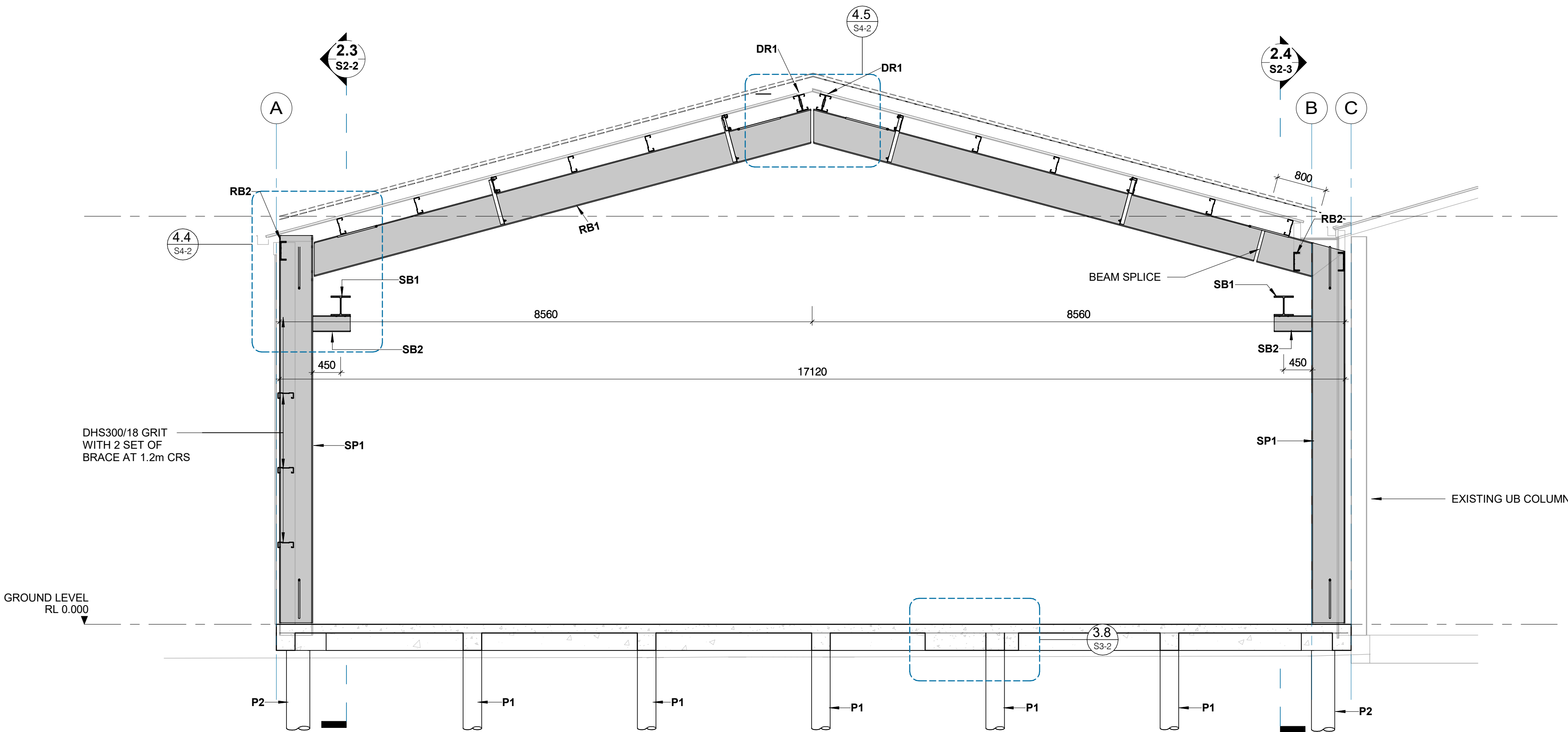
74 PATIKI RD, AVONDALE

Drawing Title
ROOF FRAMING PLAN

Engineer: AF	Drawn: DT	Date: MAY'25
Checked: AF	Drg Size: A1	Scales: As indicated
Job No.: SHL6538	Drg. No.: S1-2	Rev: A



2.1 SECTION (GRID 9)
S1-1 1 : 50 @A1 1 : 100 @A3



2.2 SECTION (GRID 5)
S1-1 1 : 50 @A1 1 : 100 @A3

NOTES

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- REFER TO COVER PAGE & SHEET S0-1 FOR STEEL PROTECTION COATING REQUIREMENTS.
- REFER TO SHEETS S0-1 & S0-2 FOR TYPICAL CONSTRUCTION NOTES & DETAILS.

STRUCTURAL LEGEND

STRUCTURAL FOUNDATION	
P1	300 SED H5 'UGLY' TANAPOLE DRIVEN TO REFUSAL. 8.0m MINIMUM BELOW GROUND BEAM INVERT. TEST PILES ARE REQUIRED TO VERIFY PILE LENGTH. 10mm SET PER BLOW FOR 5 CONSECUTIVE BLOWS (HAMMER 500 kG, DROP 1.0m)
P2	375 SED H5 'UGLY' TANAPOLE DRIVEN TO REFUSAL. 10m MINIMUM BELOW GROUND BEAM INVERT. TEST PILES ARE REQUIRED TO VERIFY PILE LENGTH. 10mm SET PER BLOW FOR 5 CONSECUTIVE BLOWS (HAMMER 500 kG, DROP 1.0m)
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STRUCTURAL SLAB	
S1	140mm THK 30MPa CONCRETE RAFT SLAB REINFORCED WITH H12-150 CRS EW TOP AND H12-150 CRS BOT ACROSS FLOOR (SHORT SPAN) AND H12-300 CRS BOT ALONG SLAB (LONG SPAN)
STRUCTURAL COLUMN	
SP1	530UB93
SP2	250 PFC
STRUCTURAL FRAMING	
GB1	300W x 420D 30MPa CONCRETE GROUND BEAM. REINFORCED WITH 3-H16 TOP & BOT PLUS ADDITIONAL 2-H16 x 2.0m LONG OVER ALL PILE AND RH10 STIRRUPS AT 150 CRS. 50mm TYP COVER
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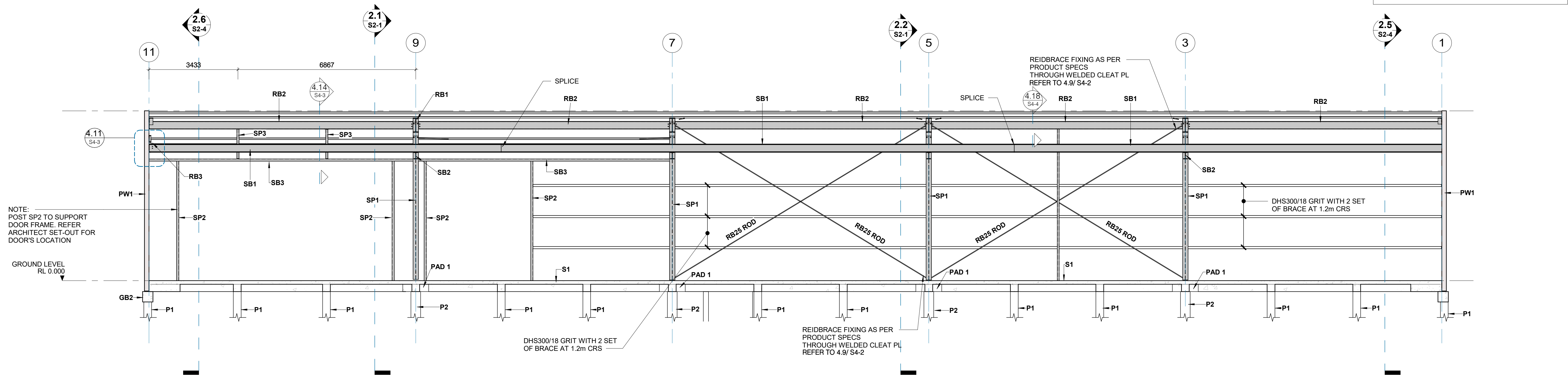
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Project
WAREHOUSE EXTENSION

74 PATIKI RD, AVONDALE

Drawing Title
SECTIONS & ELEVATIONS

Engineer: AF	Drawn: DT	Date: MAY'25
Checked: AF	Drg Size: A1	Scales: 1 : 50
Job No.: SHL6538	Drg. No.: S2-1	Rev: A



NOTE:
POST SP2 TO SUPPORT
DOOR FRAME. REFER
ARCHITECT SET-OUT FOR
DOOR'S LOCATION

GROUND LEVEL
RL 0.000

2.3 SECTION (GRID A)
S1-1 1: 75 @A1 1: 150 @A3

- NOTES
1.

REFER TO COVER PAGE FOR MANDATORY ENGINEER INSPECTIONS.
2.

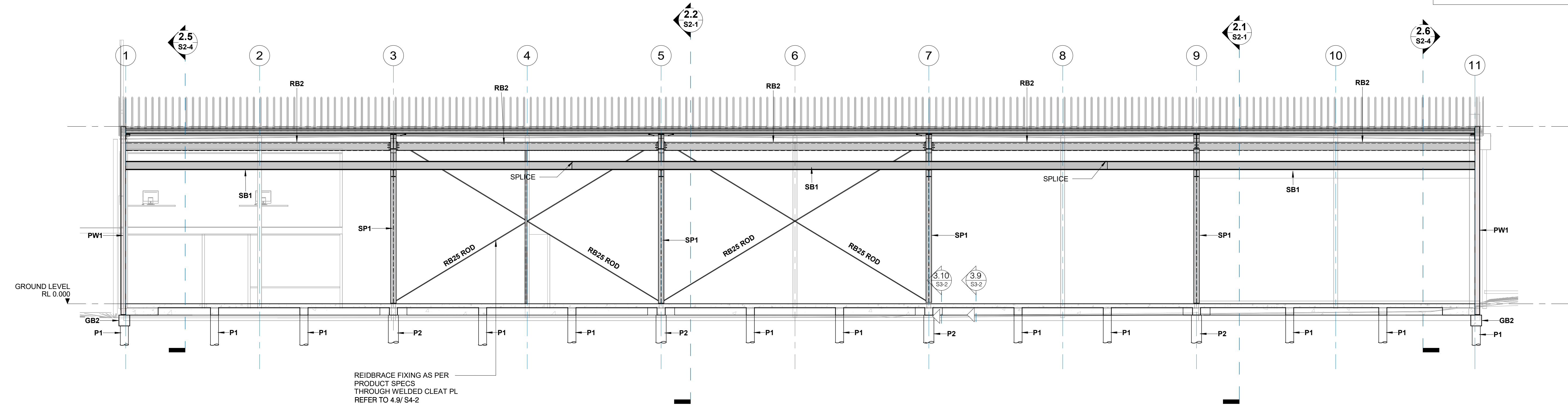
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STRUCTURAL WALL	
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STRUCTURAL SLAB	
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S3	150mm THK 30MPa CONCRETE SLAB ON GRADE. REINFORCED WITH SE92 MESH, ON SAND BLINDING ON 100 MIN COMPACTED HARDFILL

STRUCTURAL LEGEND	
STRUCTURAL COLUMN	
SP1	530UB93
SP2	250 PFC
SP3	89x6 SHS POST
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A	2025.08.11	BUILDING CONSENT
Rev	Date	Description
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WAREHOUSE EXTENSION		
74 PATIKI RD, AVONDALE		
Drawing Title		
SECTIONS & ELEVATIONS		
Engineer: AF	Drawn: DT	Date: MAY'25
Checked: AF	Drg Size: A1	Scales: 1 : 75
Job No.: SHL6538	Drg. No.: S2-2	Rev: A



- NOTES
1.

REFER TO COVER PAGE FOR MANDATORY ENGINEER INSPECTIONS.
2.

REFER TO COVER PAGE & SHEET S0-1 FOR STEEL PROTECTION COATING REQUIREMENTS.
3.

REFER TO SHEETS S0-1 & S0-2 FOR TYPICAL CONSTRUCTION NOTES & DETAILS.

2.4 SECTION (GRID B)

S1-1 1 : 75 @A1 1 : 150 @A3

STRUCTURAL LEGEND	
STRUCTURAL FOUNDATION	
P1	300 SED H5 'UGLY' TANAPOLE DRIVEN TO REFUSAL, 8.0m MINIMUM BELOW GROUND BEAM INVERT. TEST PILES ARE REQUIRED TO VERIFY POLE LENGTH. 10mm SET PER BLOW FOR 5 CONSECUTIVE BLOWS (HAMMER 500 KG, DROP 1.0m)
P2	375 SED H5 'UGLY' TANAPOLE DRIVEN TO REFUSAL, 10m MINIMUM BELOW GROUND BEAM INVERT. TEST PILES ARE REQUIRED TO VERIFY POLE LENGTH. 10mm SET PER BLOW FOR 5 CONSECUTIVE BLOWS (HAMMER 500 KG, DROP 1.0m)
PAD 1	1000x800x420D 30MPa CONCRETE PAD, REINFORCED WITH H12-150 CRS EW TOP & BOT
STRUCTURAL WALL	
PW1	175 THK 40MPa CONCRETE PRECAST WALL. REINFORCED WITH D12-100 CRS VERTICAL AND D12-250 CRS HORIZONTAL
STRUCTURAL SLAB	
S1	140mm THK 30MPa CONCRETE RAFT SLAB REINFORCED WITH H12-150 CRS EW TOP AND H12-150 CRS BOT ACROSS FLOOR (SHORT SPAN) AND H12-300 CRS BOT ALONG SLAB (LONG SPAN)
STRUCTURAL COLUMN	
SP1	530UB93
SP4	75x6 SHS
STRUCTURAL FRAMING	
GB1	300W x 420D 30MPa CONCRETE GROUND BEAM. REINFORCED WITH 3-H16 TOP & BOT PLUS ADDITIONAL 2-H16 x 2.0m LONG OVER ALL PILE AND RH10 STIRRUPS AT 150 CRS, 50mm TYP COVER
GB2	400W x 400D 30MPa CONCRETE GROUND BEAM. REINFORCED WITH 4-H16 TOP & BOT AND RH10 STIRRUPS AT 150 CRS, 75mm TYP COVER
R1	DHS250/15 WITH 2 SET OF BRACE AT 1200 CRS
RB1	530UB93
RB2	300PFC
RB4	150x10 EA
RB25 ROD	25DIA REIDBRACE ROD
SB1	310UC118
SB2	250UB37

A	2025.08.11	BUILDING CONSENT
Rev	Date	Description
REVISIONS		

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Sullivan Hall

CHARTERED ENGINEERS

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Project

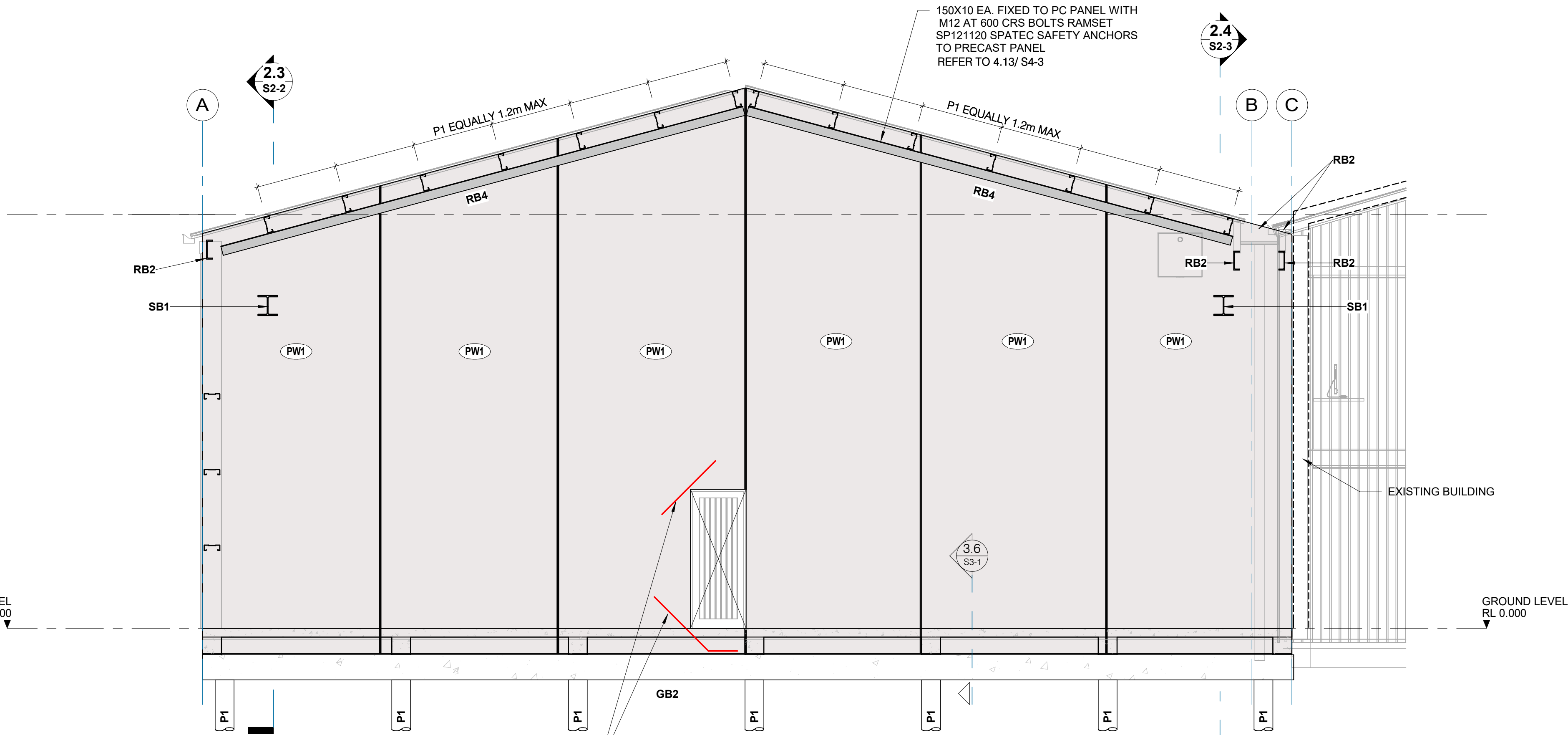
WAREHOUSE EXTENSION

74 PATIKI RD, AVONDALE

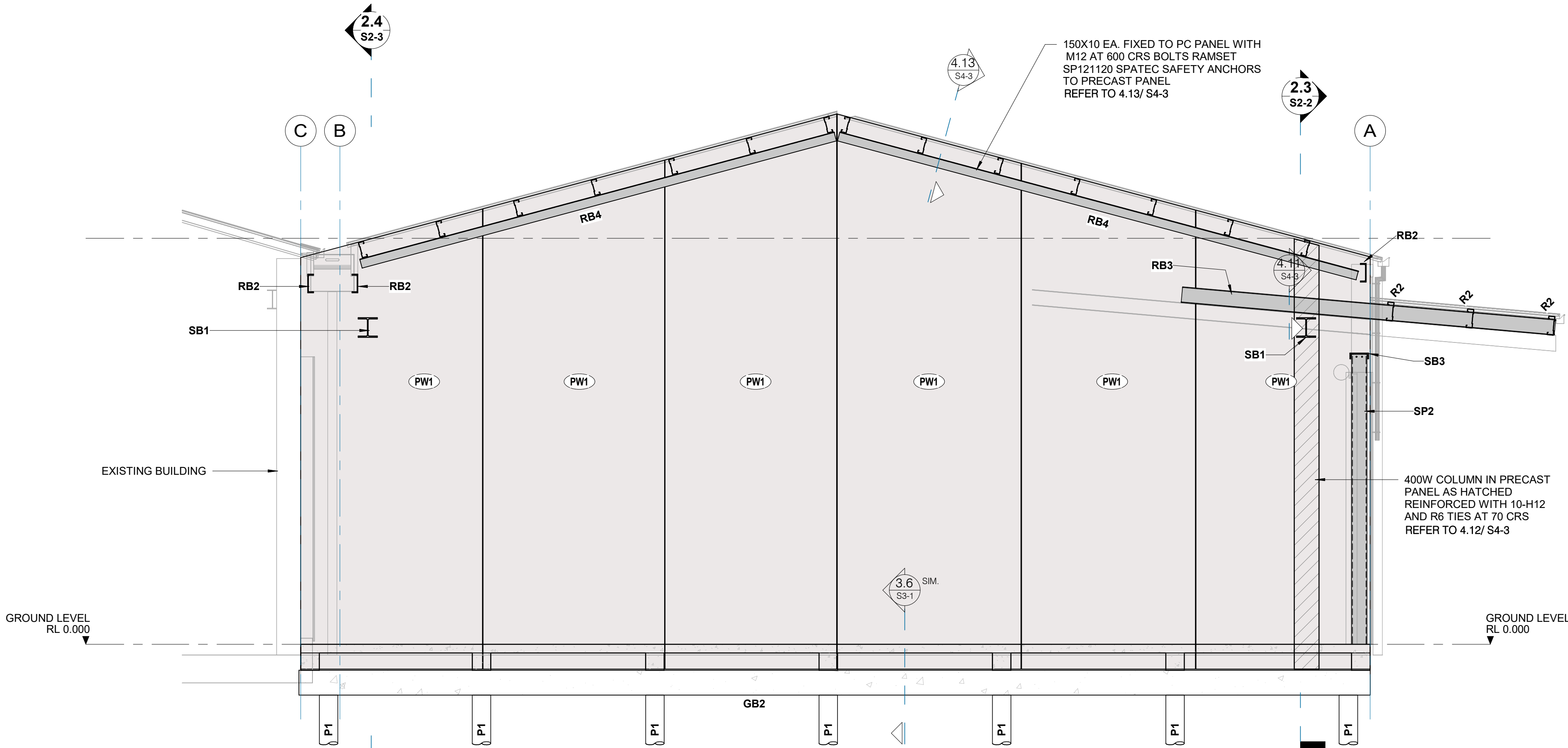
Drawing Title

SECTIONS & ELEVATIONS

Engineer: AF	Drawn: DT	Date: MAY'25
Checked: AF	Drg Size: A1	Scales: 1 : 75
Job No.: SHL6538	Drg. No.: S2-3	Rev: A



2.5 PANEL ELEVATION (GRID 1)
S1-1 1 : 50 @A1 1 : 100 @A3



2.6 PANEL ELEVATION (GRID 11)
S1-1 1 : 50 @A1 1 : 100 @A3

NOTES

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- REFER TO SHEETS S0-1 & S0-2 FOR TYPICAL CONSTRUCTION NOTES & DETAILS.

STRUCTURAL LEGEND

STRUCTURAL FOUNDATION	
P1	300 SED H5 'UGLY' TANAPOLE DRIVEN TO REFUSAL, 8.0m MINIMUM BELOW GROUND BEAM INVERT. TEST PILES ARE REQUIRED TO VERIFY POLE LENGTH. 10mm SET PER BLOW FOR 5 CONSECUTIVE BLOWS (HAMMER 500 KG, DROP 1.0m)
STRUCTURAL WALL	
PW1	175 THK 40MPa CONCRETE PRECAST WALL. REINFORCED WITH D12-100 CRS VERTICAL AND D12-250 CRS HORIZONTAL
STRUCTURAL SLAB	
S1	140mm THK 30MPa CONCRETE RAFT SLAB REINFORCED WITH H12-150 CRS EW TOP AND H12-150 CRS BOT ACROSS FLOOR (SHORT SPAN) AND H12-300 CRS BOT ALONG SLAB (LONG SPAN)
STRUCTURAL COLUMN	
SP2	250 PFC
STRUCTURAL FRAMING	
GB1	300W x 420D 30MPa CONCRETE GROUND BEAM. REINFORCED WITH 3-H16 TOP & BOT PLUS ADDITIONAL 2-H16 x 2.0m LONG OVER ALL PILE AND RH10 STIRRUPS AT 150 CRS. 50mm TYP COVER
GB2	400W x 400D 30MPa CONCRETE GROUND BEAM. REINFORCED WITH 4-H16 TOP & BOT AND RH10 STIRRUPS AT 150 CRS. 75mm TYP COVER
R1	DHS250/15 WITH 2 SET OF BRACE AT 1200 CRS
R2	DHS300/18 AT 1200 CRS
RB2	300PFC
RB3	250UB37
RB4	150x10 EA
SB1	310UC118
SB3	300 PFC ON FLAT

Rev	Date	Description
A	2025.08.11	BUILDING CONSENT
REVISIONS		

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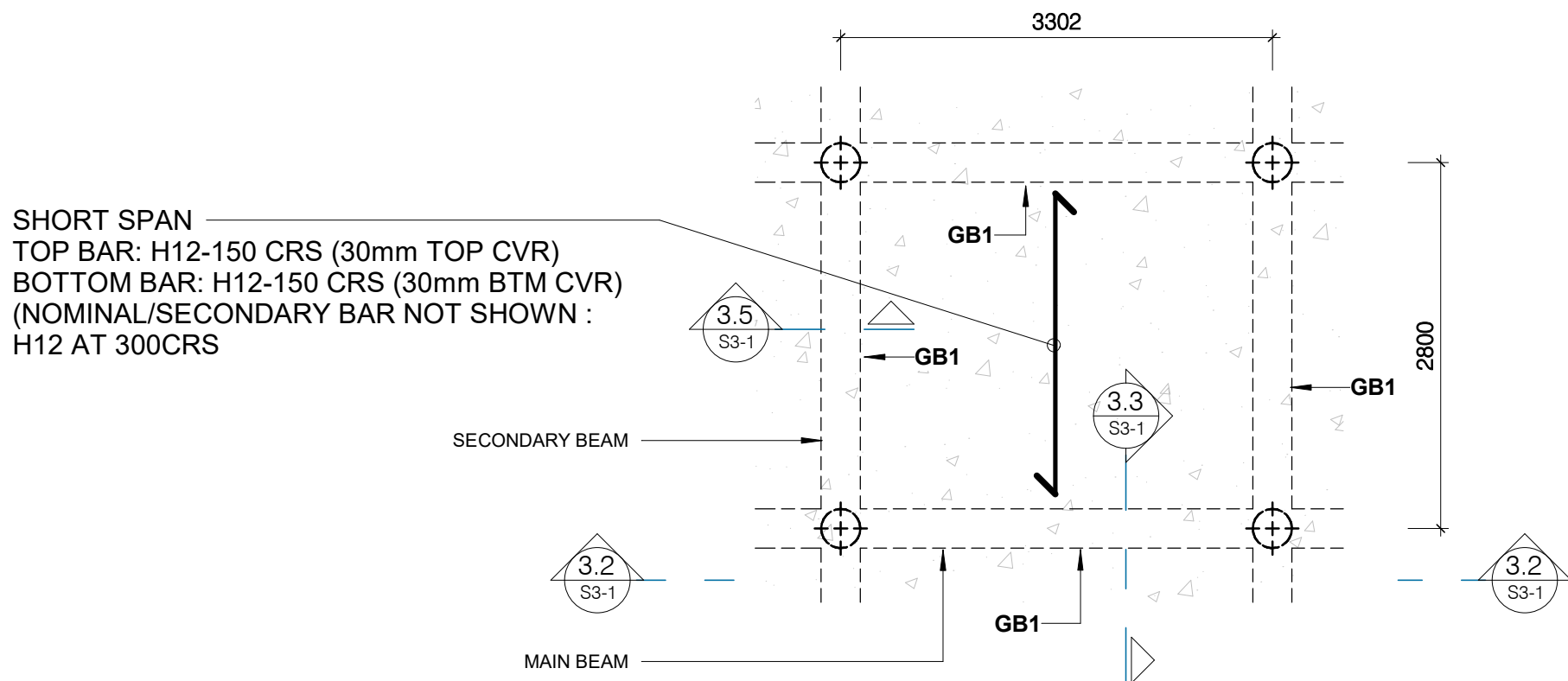
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Project
WAREHOUSE EXTENSION

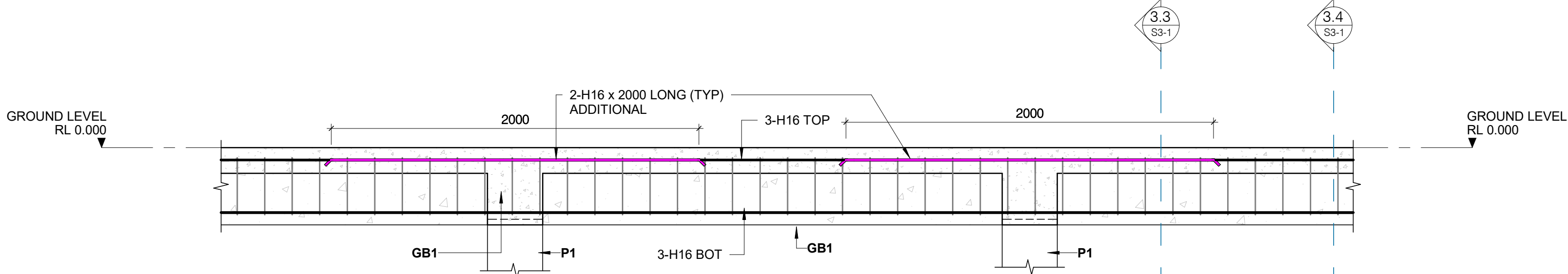
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Drawing Title
SECTIONS & ELEVATIONS

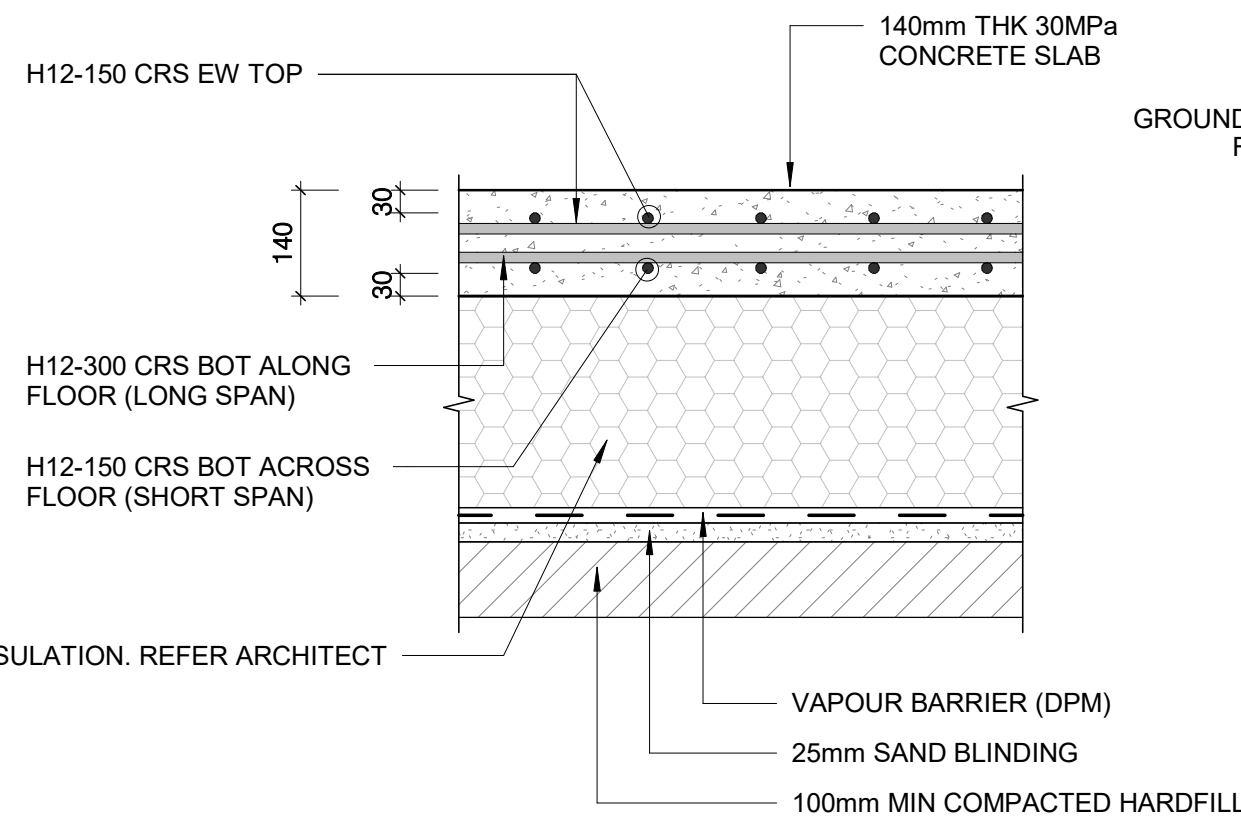
Engineer: AF	Drawn: DT	Date: MAY'25
Checked: AF	Drg Size: A1	Scales: 1 : 50
Job No.: SHL6538	Drg. No.: S2-4	Rev: A



3.0 DETAIL (PLAN VIEW)
S1-1 1 : 50 @A1 1 : 40 @A3

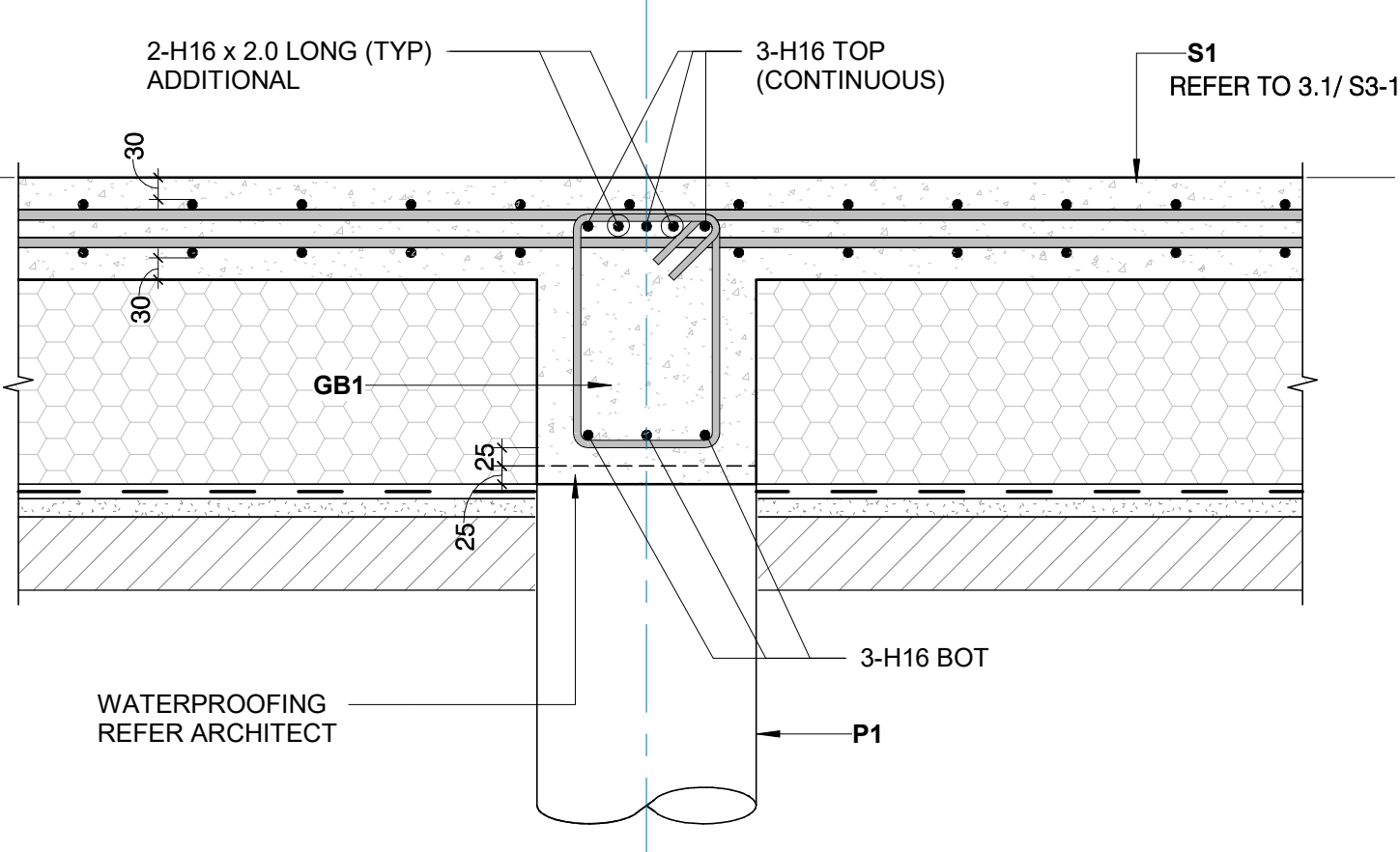


3.2 DETAIL
S3-1 1 : 20 @A1 1 : 40 @A3

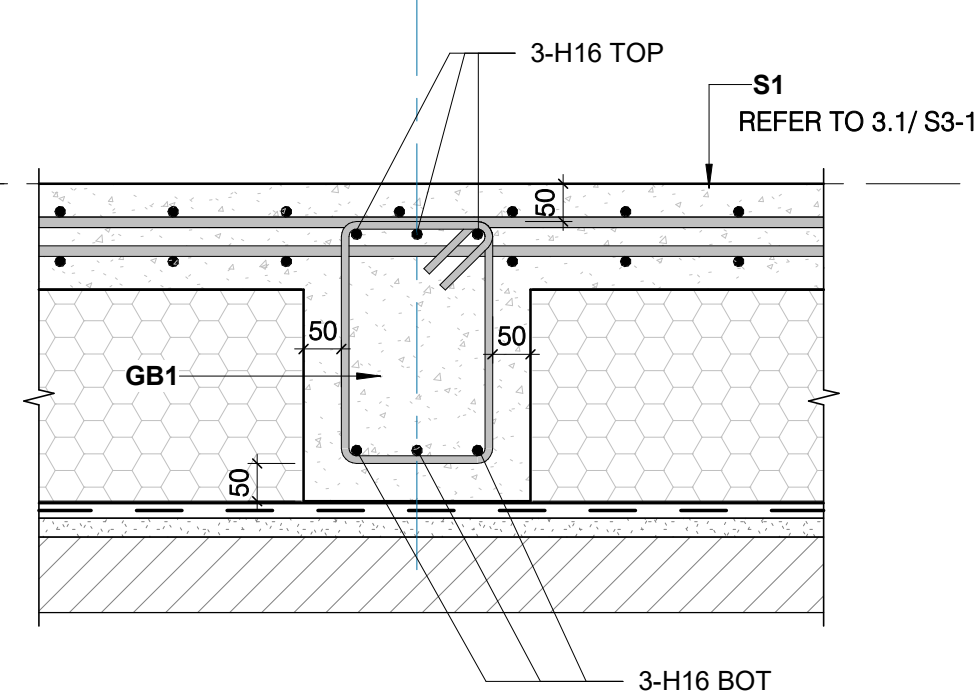


S1 SLAB TYP DETAIL

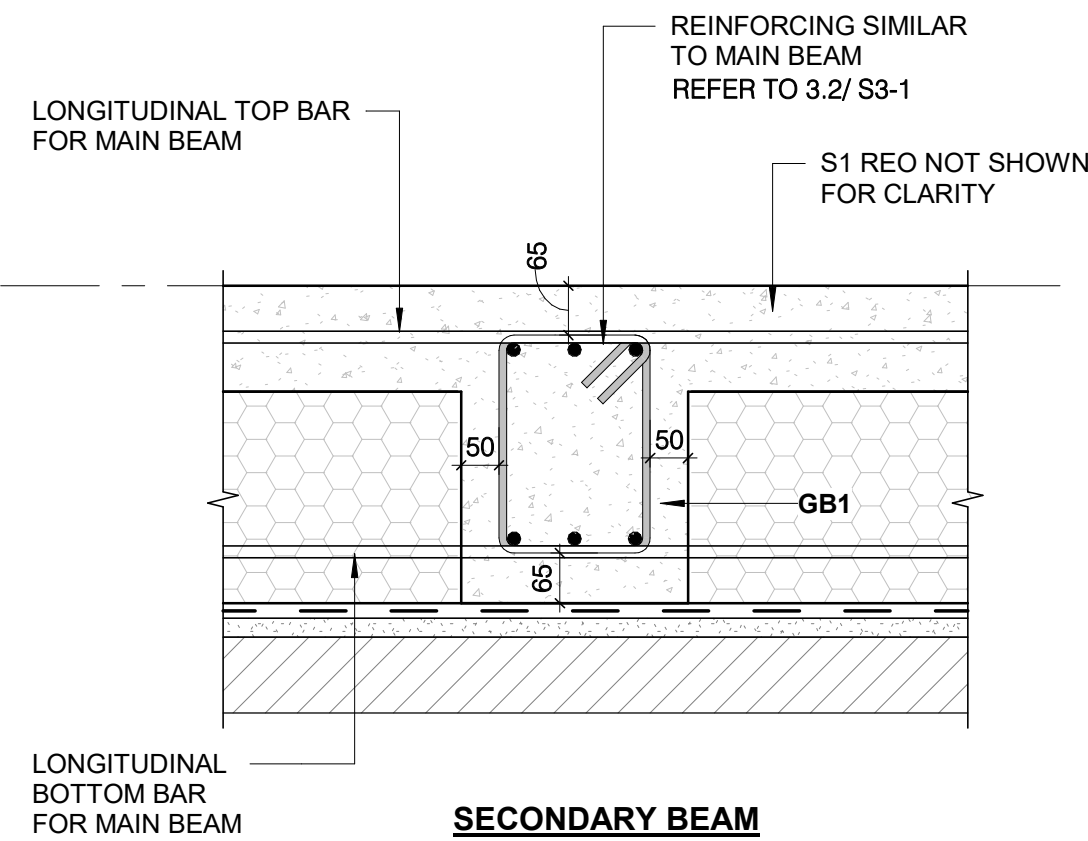
3.1 DETAIL
S1-1 1 : 10 @A1 1 : 20 @A3



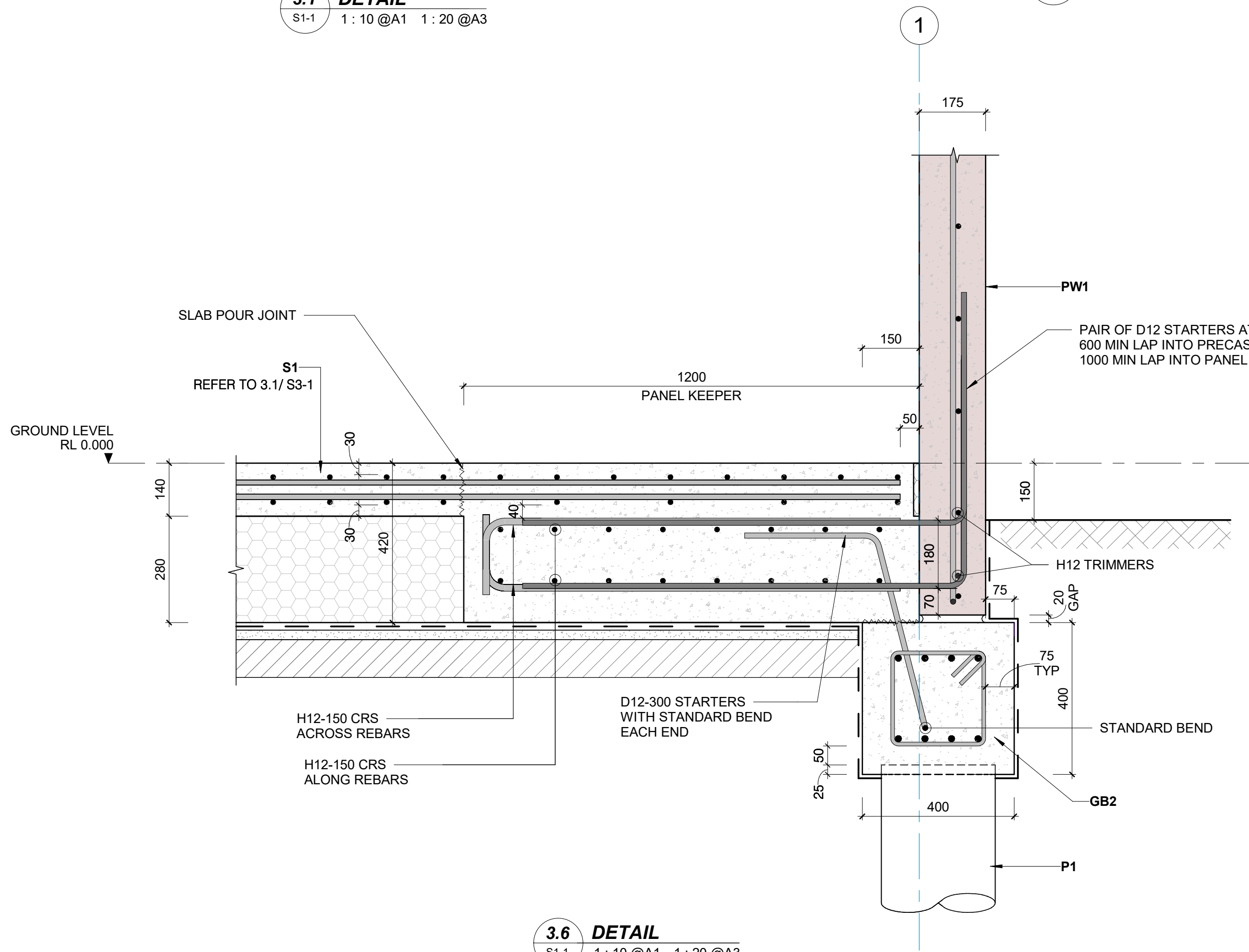
3.3 DETAIL
S3-1 1 : 10 @A1 1 : 20 @A3



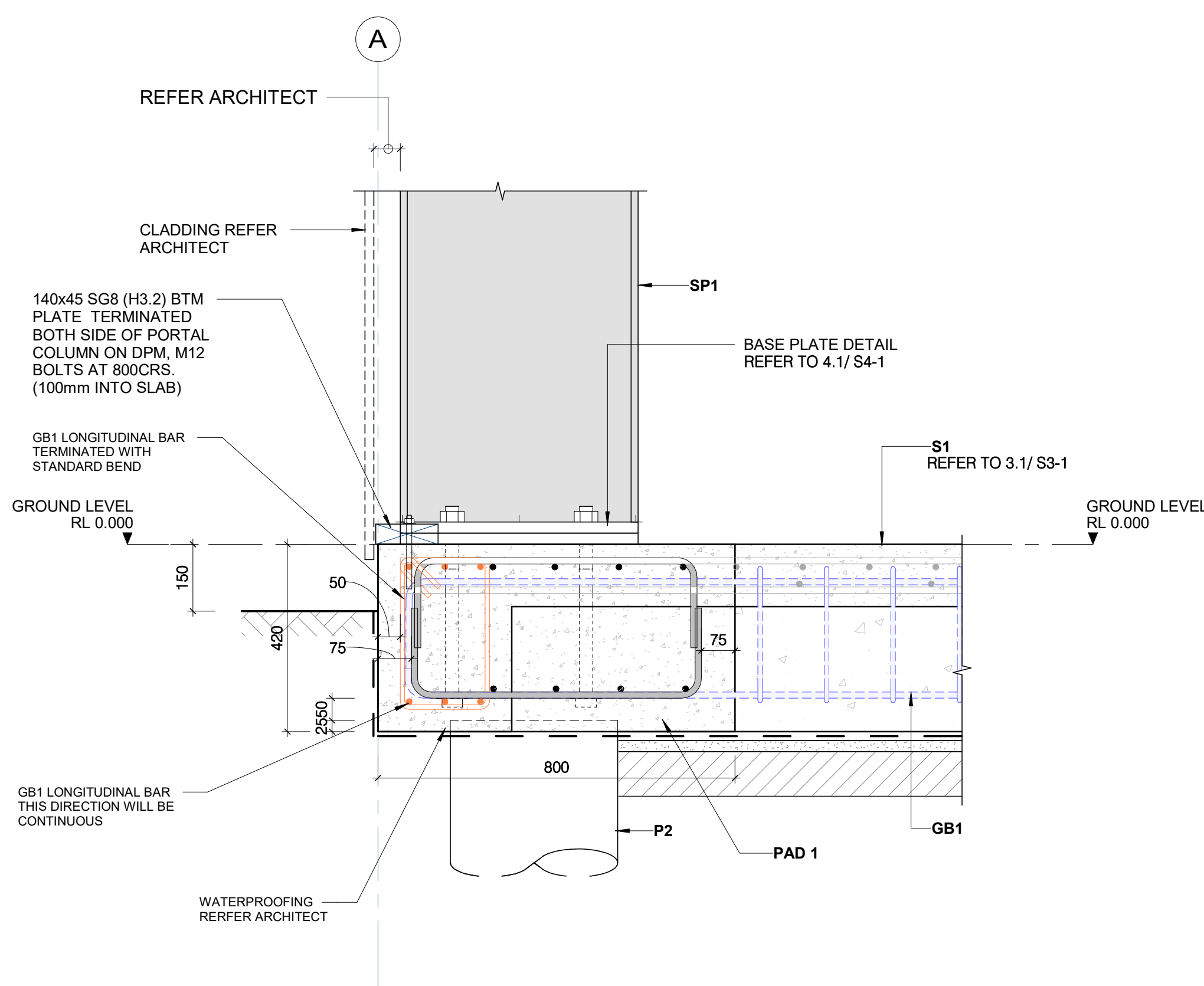
3.4 DETAIL
S3-1 1 : 10 @A1 1 : 20 @A3



3.5 DETAIL
S3-1 1 : 10 @A1 1 : 20 @A3



3.6 DETAIL
S1-1 1 : 10 @A1 1 : 20 @A3



3.7 DETAIL
S1-1 1 : 10 @A1 1 : 20 @A3

NOTES

- REFER TO COVER PAGE FOR MANDATORY ENGINEER INSPECTIONS.
- REFER TO COVER PAGE & SHEET S0-1 FOR STEEL PROTECTION COATING REQUIREMENTS.
- REFER TO SHEETS S0-1 & S0-2 FOR TYPICAL CONSTRUCTION NOTES & DETAILS.

STRUCTURAL LEGEND

STRUCTURAL FOUNDATION	
P1	300 SED H5 'UGLY' TANAPOLE DRIVEN TO REFUSAL. 8.0m MINIMUM BELOW GROUND BEAM INVERT. TEST PILES ARE REQUIRED TO VERIFY POLE LENGTH. 10mm SET PER BLOW FOR 5 CONSECUTIVE BLOWS (HAMMER 500 kG, DROP 1.0m)
P2	375 SED H5 'UGLY' TANAPOLE DRIVEN TO REFUSAL. 10m MINIMUM BELOW GROUND BEAM INVERT. TEST PILES ARE REQUIRED TO VERIFY POLE LENGTH. 10mm SET PER BLOW FOR 5 CONSECUTIVE BLOWS (HAMMER 500 kG, DROP 1.0m)
PAD 1	1000x800x420D 30MPa CONCRETE PAD. REINFORCED WITH H12-150 CRS EW TOP & BOT
STRUCTURAL WALL	
PW1	175 THK 40MPa CONCRETE PRECAST WALL. REINFORCED WITH D12-100 CRS VERTICAL AND D12-250 CRS HORIZONTAL
STRUCTURAL SLAB	
S1	140mm THK 30MPa CONCRETE RAFT SLAB REINFORCED WITH H12-150 CRS EW TOP AND H12-150 CRS BOT ACROSS FLOOR (SHORT SPAN) AND H12-300 CRS BOT ALONG SLAB (LONG SPAN)
STRUCTURAL COLUMN	
SP1	530UB93
STRUCTURAL FRAMING	
GB1	300W x 420D 30MPa CONCRETE GROUND BEAM. REINFORCED WITH 3-H16 TOP & BOT PLUS ADDITIONAL 2-H16 x 2.0m LONG OVER ALL PILE AND RH10 STIRRUPS AT 150 CRS. 50mm TYP COVER
GB2	400W x 400D 30MPa CONCRETE GROUND BEAM. REINFORCED WITH 4-H16 TOP & BOT AND RH10 STIRRUPS AT 150 CRS. 75mm TYP COVER

A	2025.08.11	BUILDING CONSENT
Rev	Date	Description

REVISIONS

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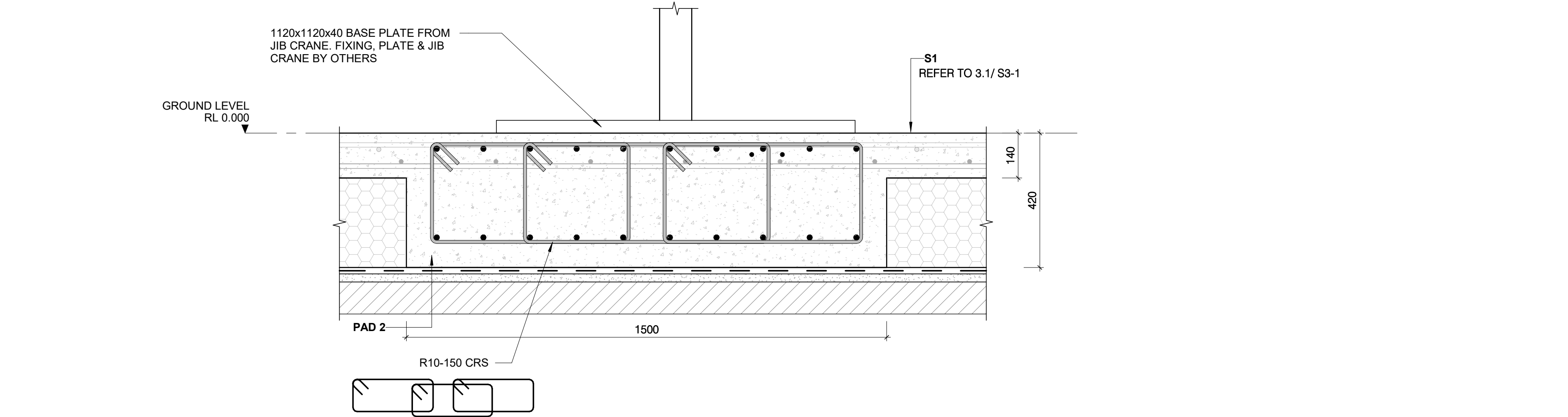
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WAREHOUSE EXTENSION

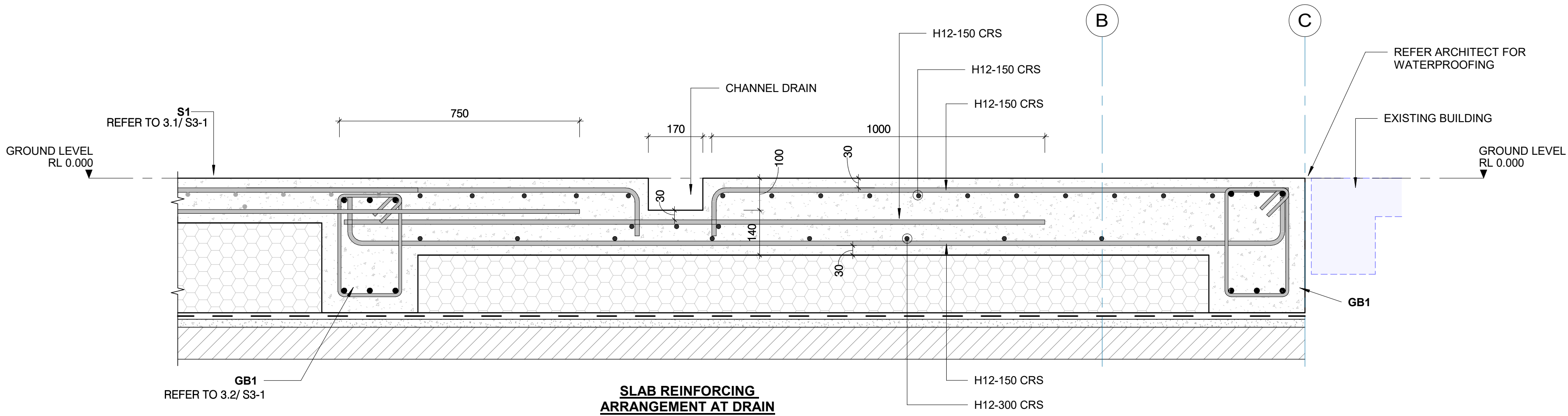
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Drawing Title
FOUNDATION DETAILS

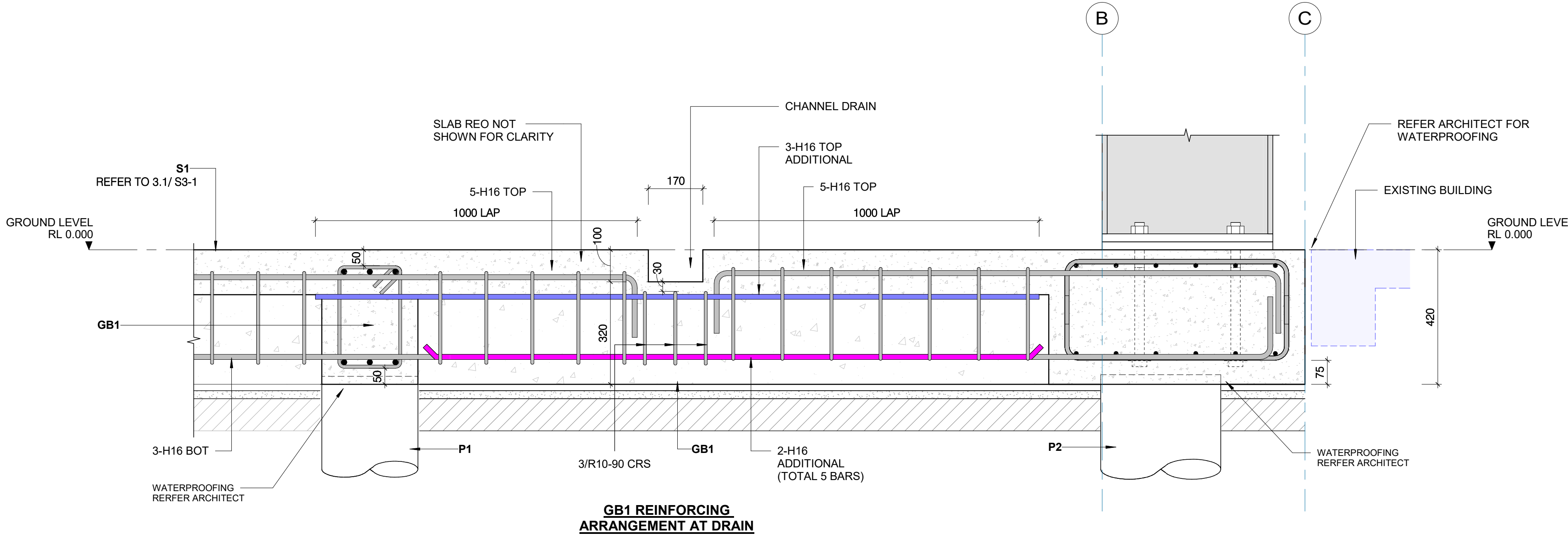
Engineer: AF	Drawn: DT	Date: MAY'25
Checked: AF	Drg Size: A1	Scales: As indicated
Job No.: SHL6538	Drg. No.: S3-1	Rev: A



3.8 DETAIL
S1-1 1: 10 @A1 1: 20 @A3



3.9 DETAIL
S1-1 1: 10 @A1 1: 20 @A3

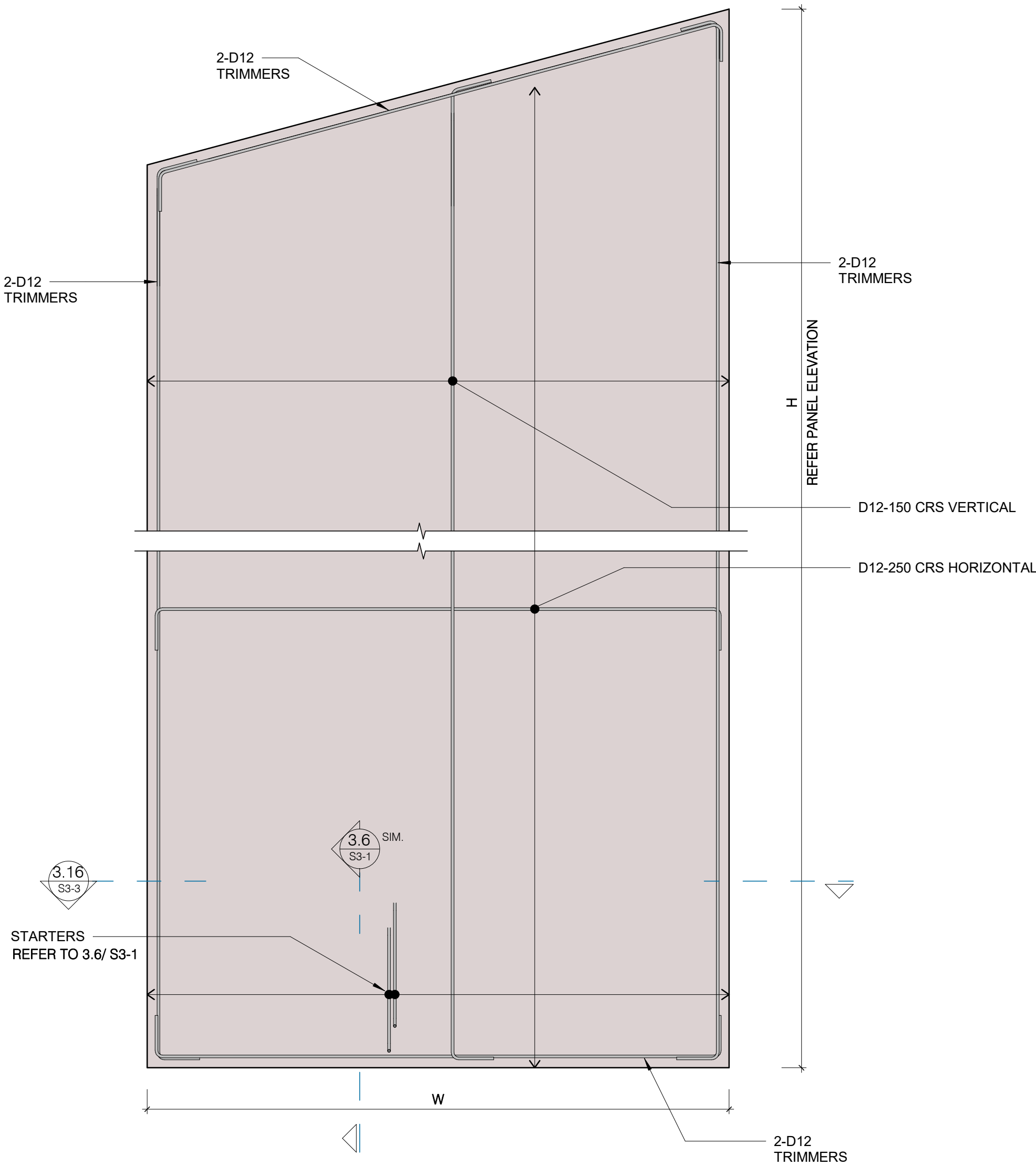


3.10 DETAIL
S1-1 1: 10 @A1 1: 20 @A3

NOTES		
1.	REFER TO COVER PAGE FOR MANDATORY ENGINEER INSPECTIONS.	
2.	REFER TO COVER PAGE & SHEET S0-1 FOR STEEL PROTECTION COATING REQUIREMENTS.	
3.	REFER TO SHEETS S0-1 & S0-2 FOR TYPICAL CONSTRUCTION NOTES & DETAILS.	

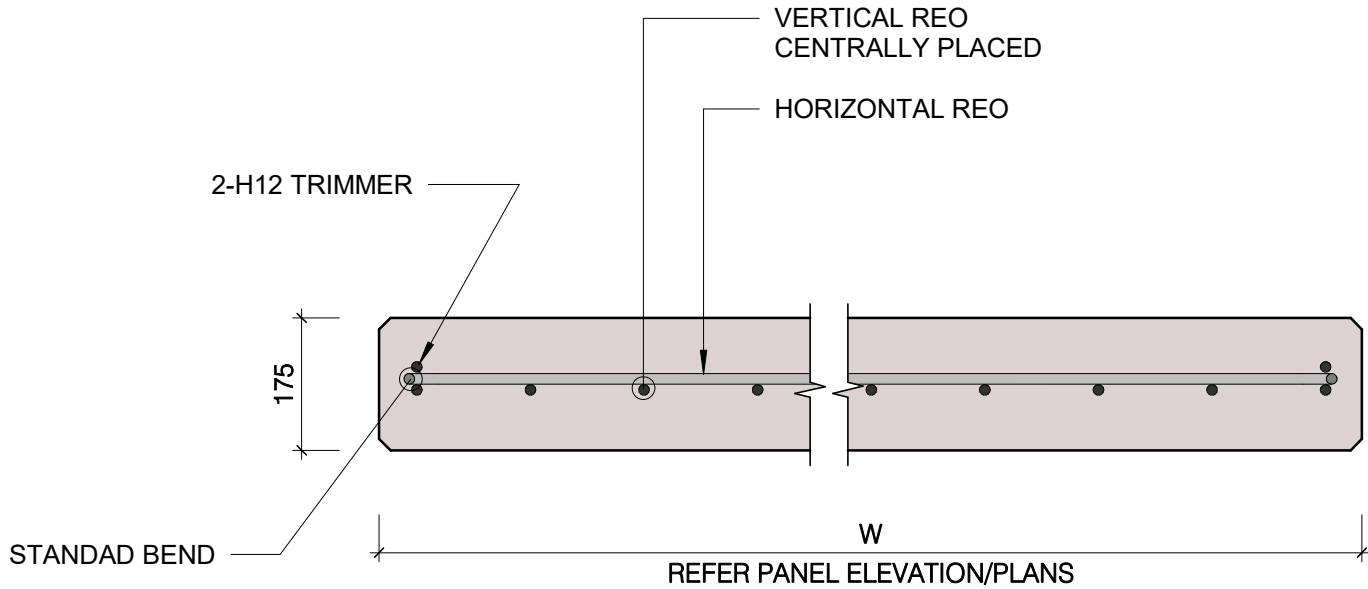
STRUCTURAL LEGEND	
STRUCTURAL FOUNDATION	
P1	300 SED H5 'UGLY' TANAPOLE DRIVEN TO REFUSAL, 8.0m MINIMUM BELOW GROUND BEAM INVERT. TEST PILES ARE REQUIRED TO VERIFY POLE LENGTH, 10mm SET PER BLOW FOR 5 CONSECUTIVE BLOWS (HAMMER 500 KG, DROP 1.0m)
P2	375 SED H5 'UGLY' TANAPOLE DRIVEN TO REFUSAL, 10m MINIMUM BELOW GROUND BEAM INVERT. TEST PILES ARE REQUIRED TO VERIFY POLE LENGTH, 10mm SET PER BLOW FOR 5 CONSECUTIVE BLOWS (HAMMER 500 KG, DROP 1.0m)
PAD 1	1000x800x420D 30MPa CONCRETE PAD, REINFORCED WITH H12-150 CRS EW TOP & BOT
PAD 2	3600x1500x420D 30MPa CONCRETE PAD, REINFORCED WITH H16-150 CRS TOP & BOT AND 6 LEGS R10-150 CRS, 75mm TYP COVER
STRUCTURAL SLAB	
S1	140mm THK 30MPa CONCRETE RAFT SLAB REINFORCED WITH H12-150 CRS EW TOP AND H12-150 CRS BOT ACROSS FLOOR (SHORT SPAN) AND H12-300 CRS BOT ALONG SLAB (LONG SPAN)
STRUCTURAL COLUMN	
SP1	530UB93
STRUCTURAL FRAMING	
GB1	300W x 420D 30MPa CONCRETE GROUND BEAM, REINFORCED WITH 3-H16 TOP & BOT PLUS ADDITIONAL 2-H16 x 2.0m LONG OVER ALL PILE AND RH10 STIRRUPS AT 150 CRS, 50mm TYP COVER

A	2025.08.11	BUILDING CONSENT
Rev	Date	Description
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Drawing Title		
FOUNDATION DETAILS		
Engineer: AF	Drawn: DT	Date: MAY'25
Checked: AF	Drg Size: A1	Scales: 1: 10
Job No.: SHL6538	Drg. No.: S3-2	Rev: A



TYP PRECAST PANEL (ELEVATION)

3.15 DETAIL
1 : 20 @A1 1 : 40 @A3



TYP PRECAST PANEL (SECTION)

3.16 DETAIL
S3-3 1 : 10 @A1 1 : 20 @A3

NOTES

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- REFER TO SHEETS S0-1 & S0-2 FOR TYPICAL CONSTRUCTION NOTES & DETAILS.

STRUCTURAL WALL SCHEDULE

TYPE MARK	DESCRIPTION
PW1	175 THK 40MPa CONCRETE PRECAST WALL. REINFORCED WITH D12-100 CRS VERTICAL AND D12-250 CRS HORIZONTAL

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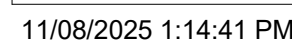
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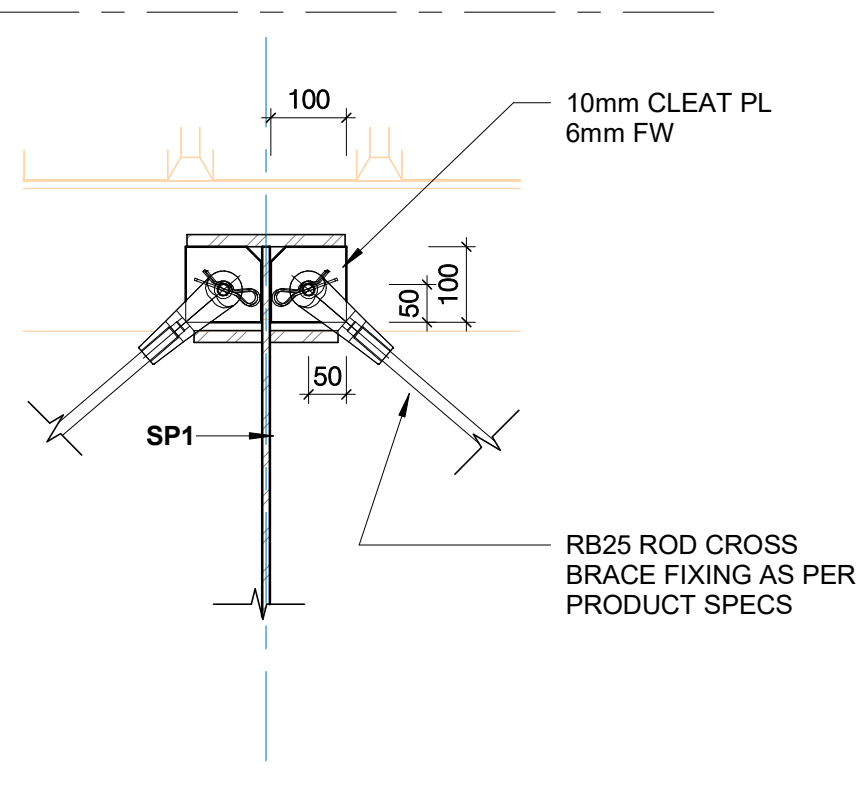
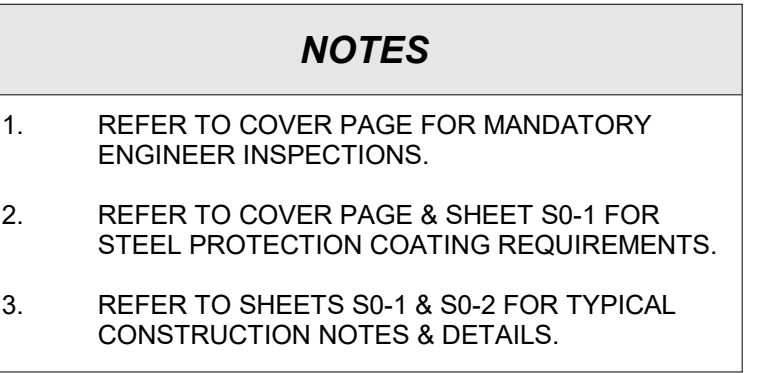
PANEL DETAILS

Engineer: AF Drawn: DT Date: MAY'25

Checked: AF Drg Size: A1 Scales: As indicated

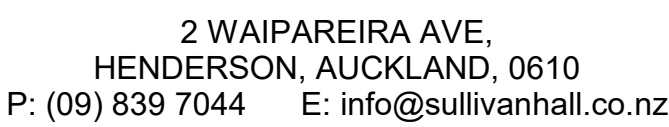
Job No.: SHL6538 Drg. No.: S3-3 Rev: A





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Drawing Title

DETAILS

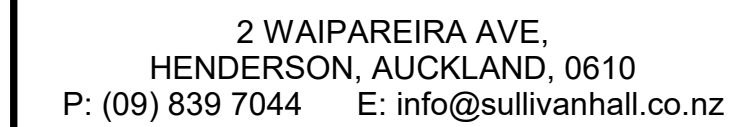
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Checked: AF	Drg Size: A1	Scales: 1 : 10
Job No.: SHL6538	Drg. No.: S4-2	Rev: A

NOTES	
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3.	REFER TO SHEETS S0-1 & S0-2 FOR TYPICAL CONSTRUCTION NOTES & DETAILS.

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Rev	Date	Description
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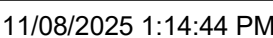
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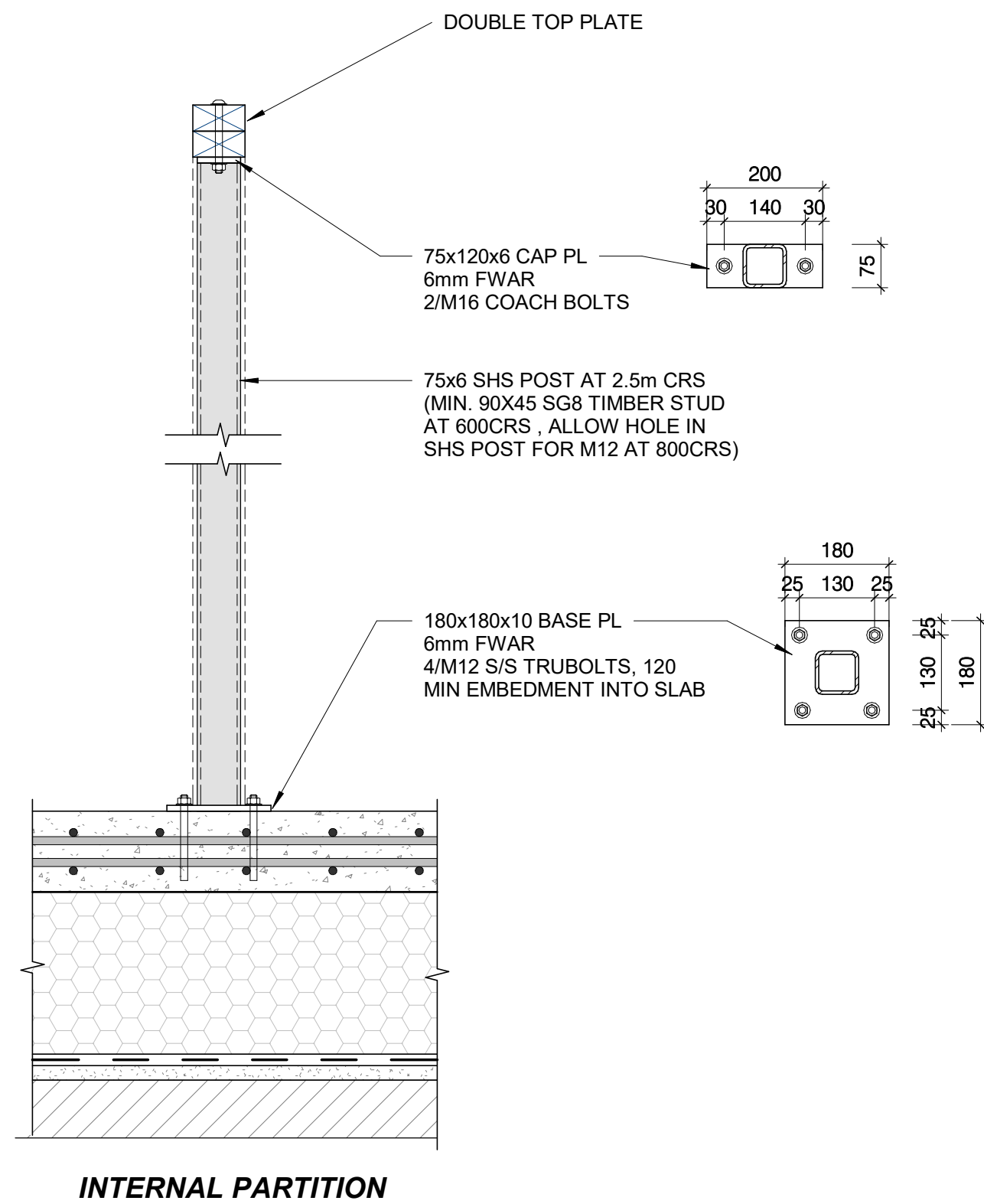
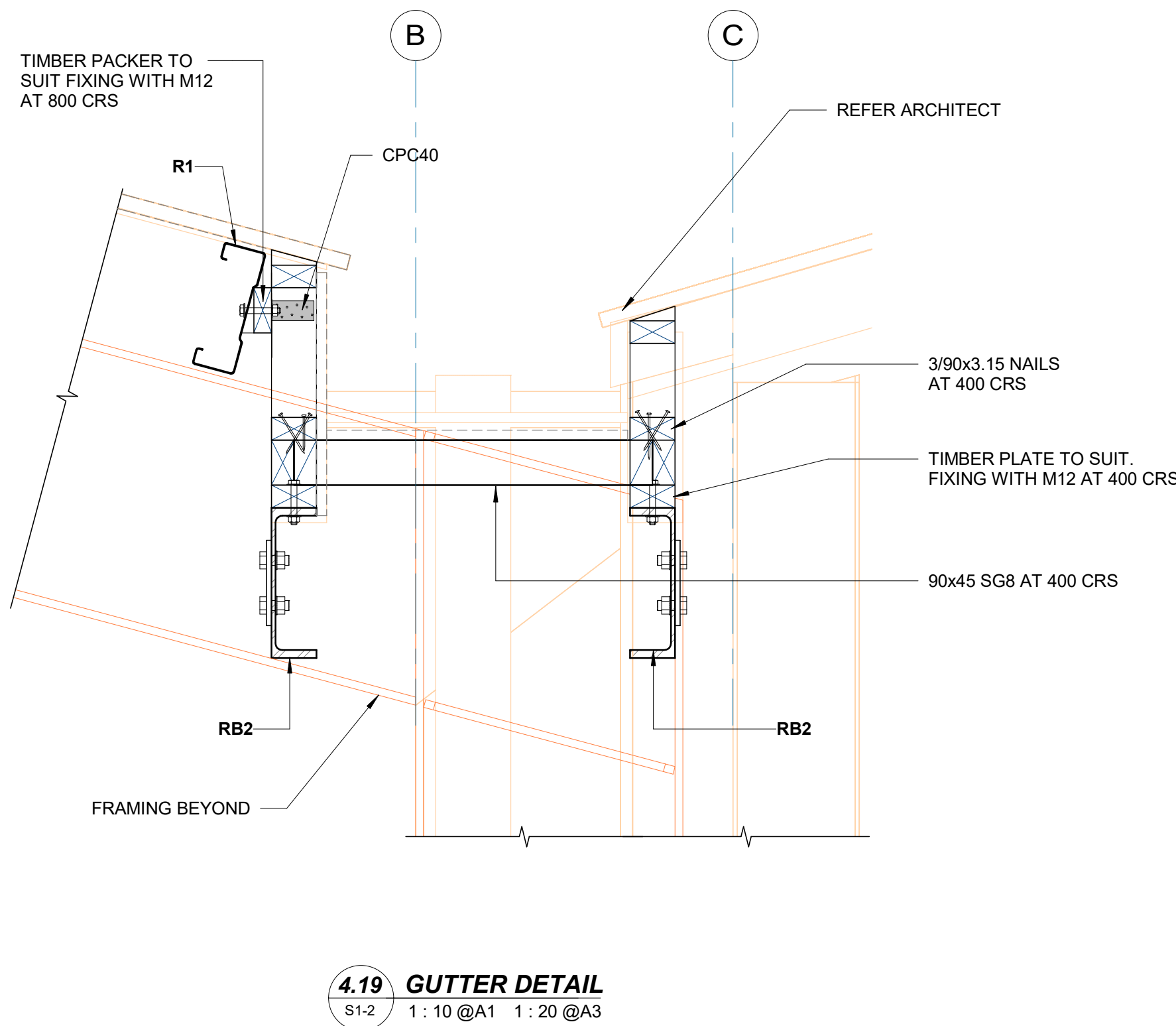
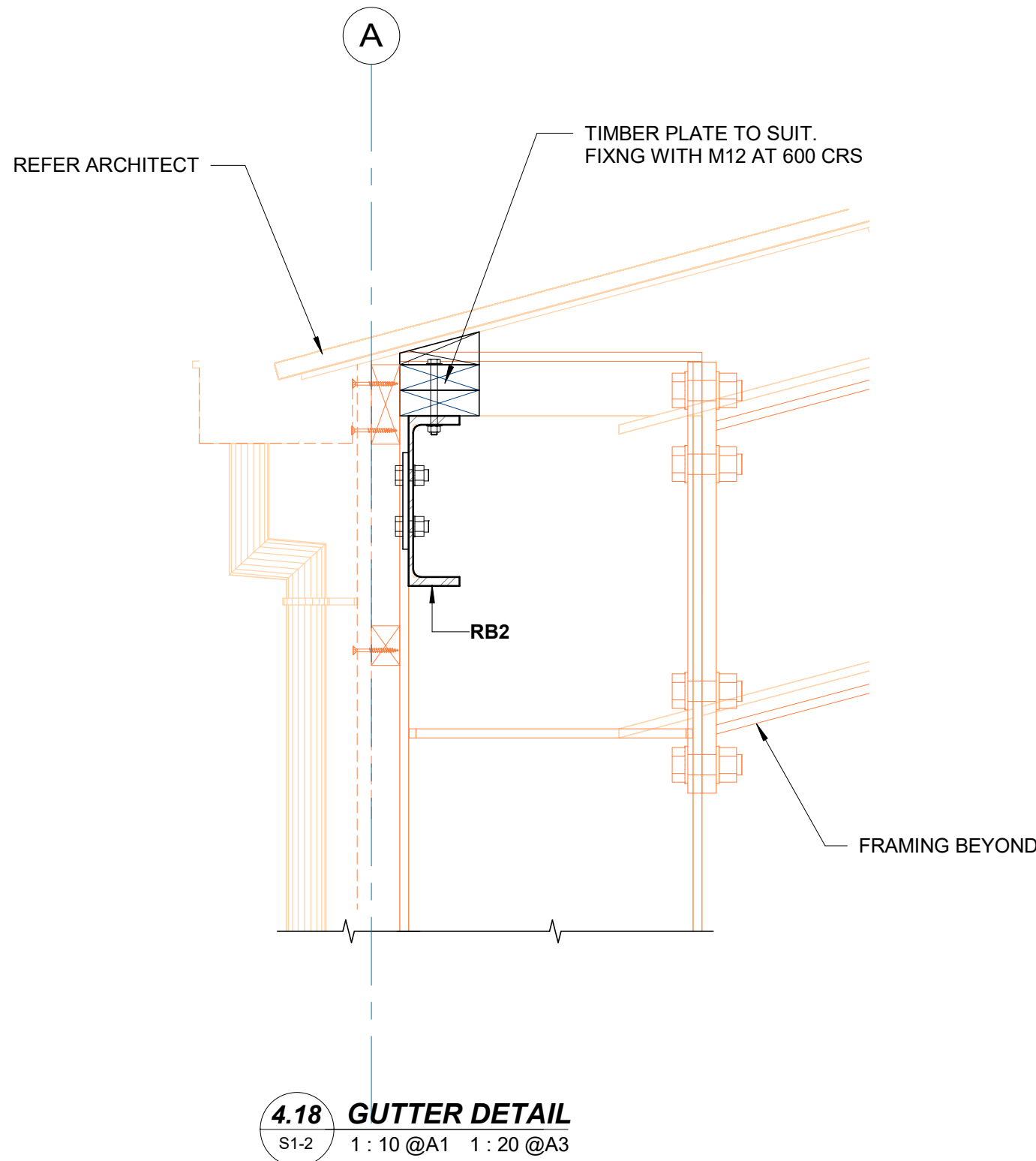
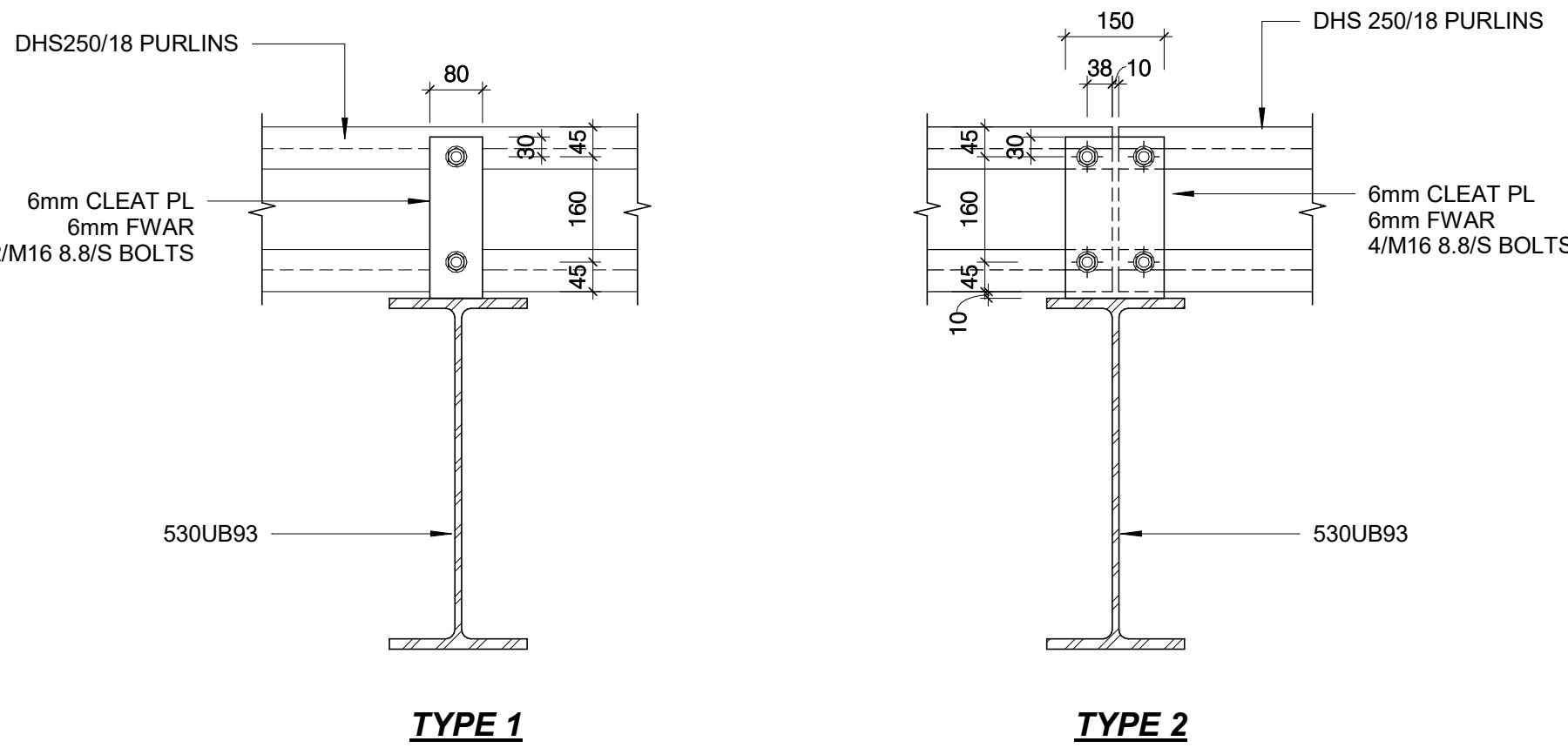
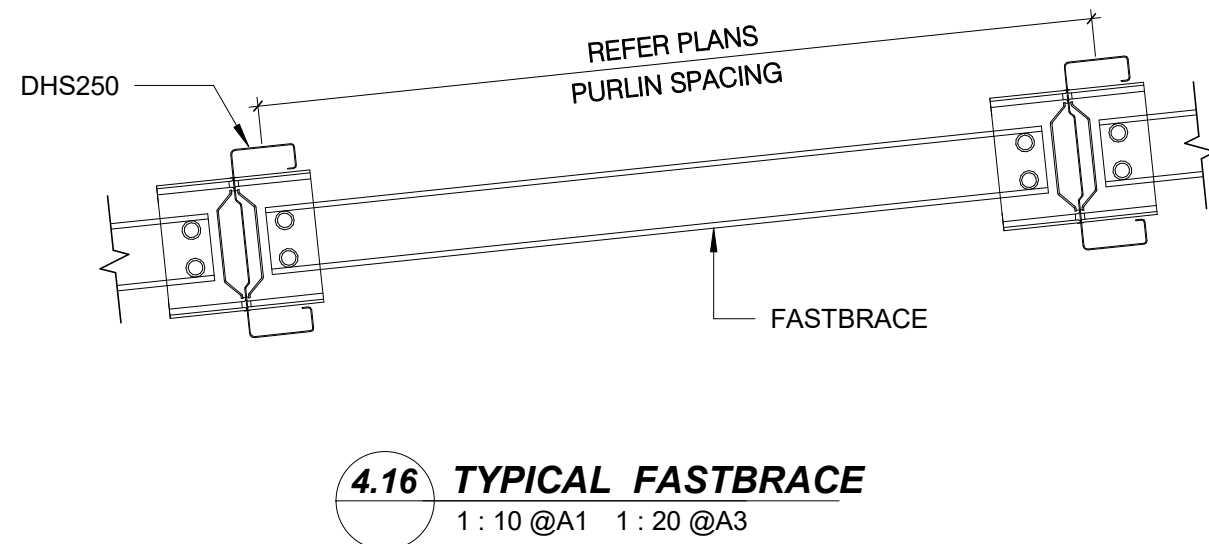
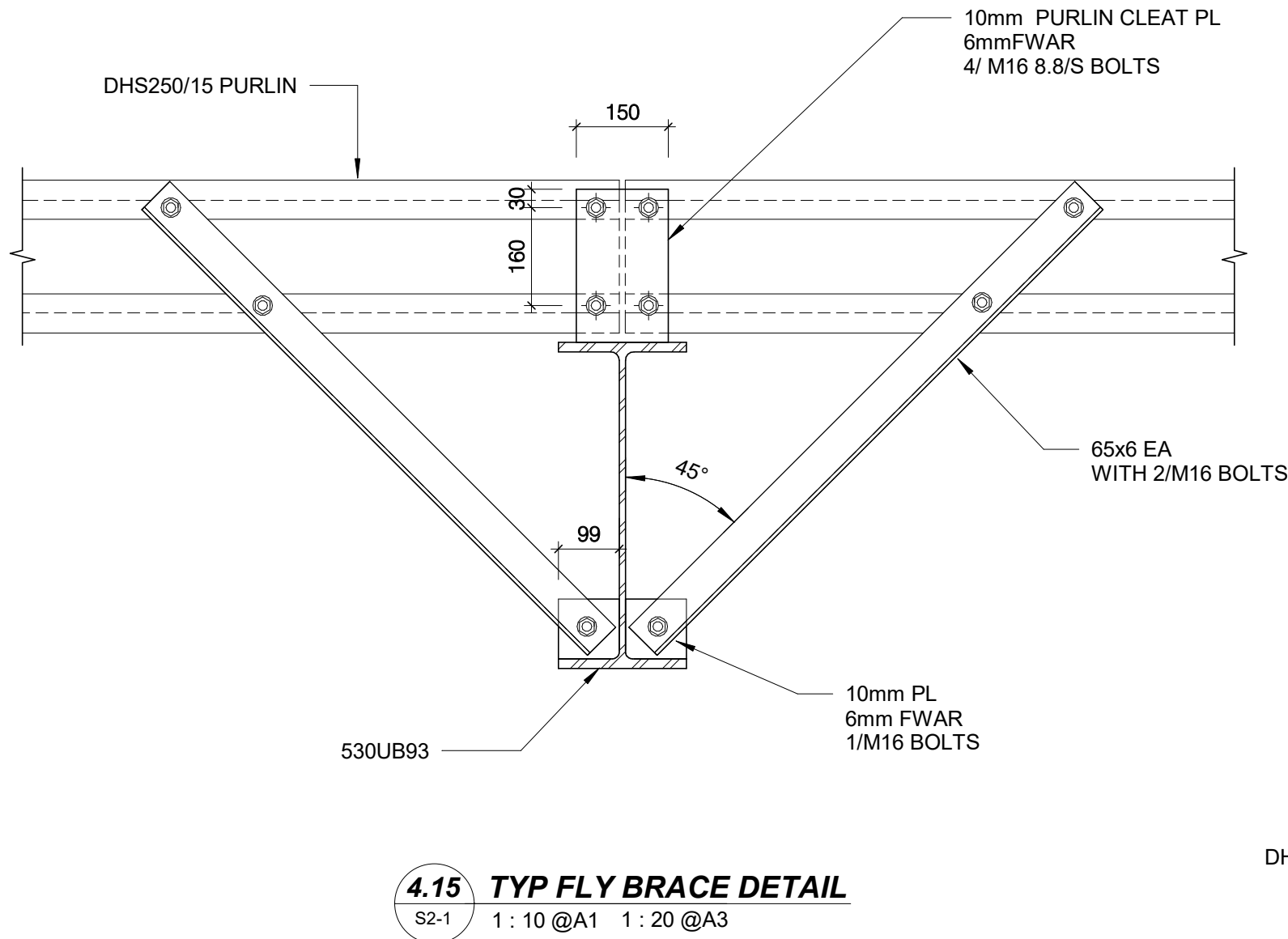
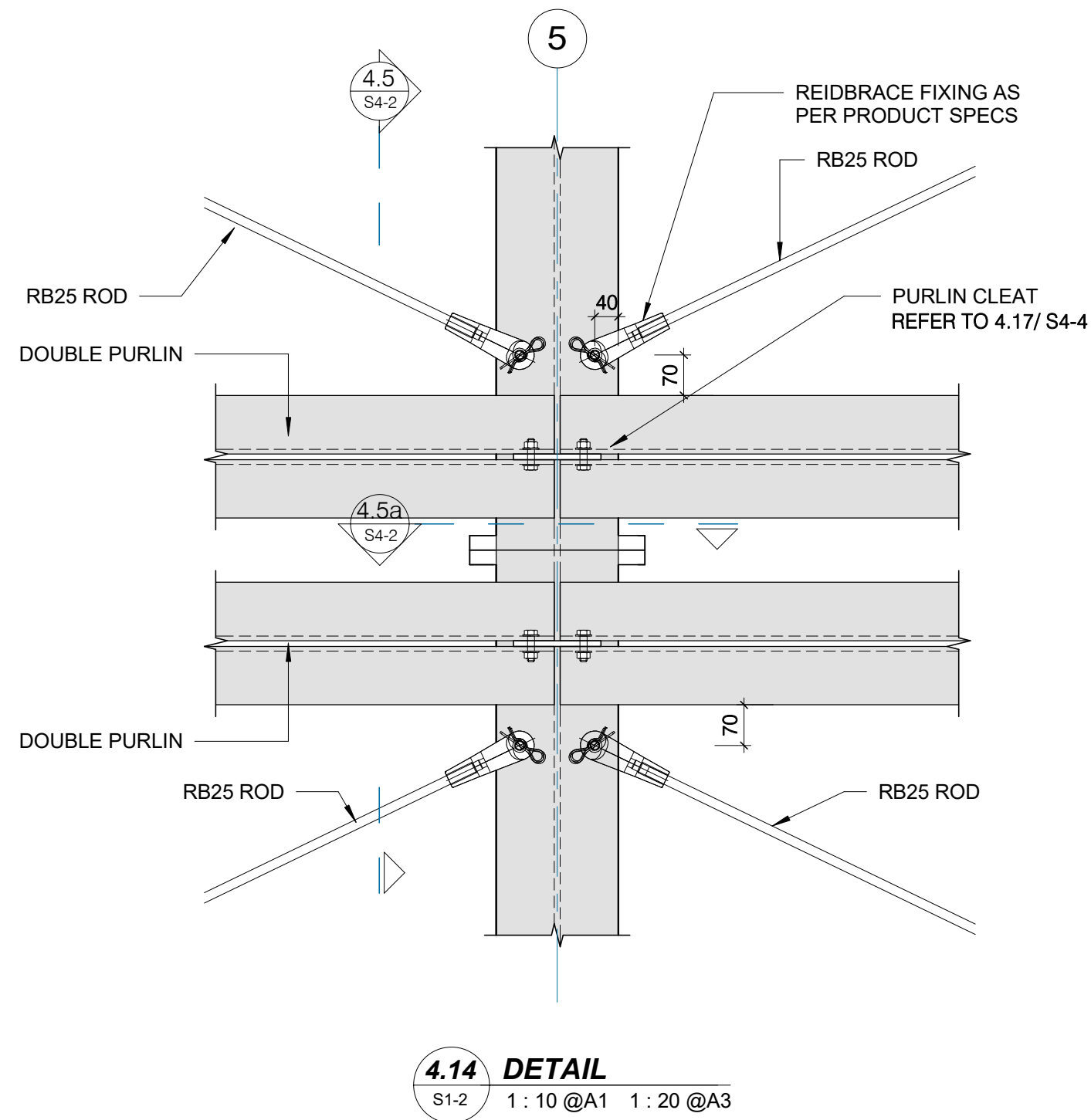
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Engineer: AF	Drawn: DT	Date: MAY'25
Checked: AF	Dwg Size: A1	Scales: 1 : 10
Job No.: SHL6538	Drg. No.: S4-3	Rev: A





NOTES	
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2.	REFER TO COVER PAGE & SHEET S0-1 FOR STEEL PROTECTION COATING REQUIREMENTS.
3.	REFER TO SHEETS S0-1 & S0-2 FOR TYPICAL CONSTRUCTION NOTES & DETAILS.

STRUCTURAL LEGEND	
STRUCTURAL COLUMN	
SP1	530UB93
STRUCTURAL FRAMING	
R1	DHS250/15 WITH 2 SET OF BRACE AT 1200 CRS
RB1	530UB93
RB2	300PFC

A	2025.08.11	BUILDING CONSENT
Rev	Date	Description
REVISIONS		
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74 PATIKI RD, AVONDALE		
Drawing Title		
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Engineer: AF	Drawn: DT	Date: MAY'25
Checked: AF	Drg Size: A1	Scales: 1 : 10
Job No.: SHL6538	Drg. No.: S4-4	Rev: A